



Serial dependence of NDARMA processes



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ABSTRACT

The NDARMA model is a discrete counterpart to the usual ARMA model, which can be applied to purely categorical processes. NDARMA processes are shown to be φ -mixing, so it is possible to find asymptotic expressions for the distribution of several types of statistics. Such asymptotic properties are useful for hypothesis testing or other inferential procedures. This is exemplified by considering the Gini index and the entropy as measures of marginal dispersion, the Pearson statistic for checking the goodness-of-fit with regard to a hypothetical marginal distribution, and several measures of signed or unsigned serial dependence. For each of these cases, the obtained asymptotic approximations are also compared to the empirically observed behavior in time series of finite length. Practical applications are illustrated by a real-data example.

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1. Introduction

A (discrete-time stochastic) process $(X_t)_{\mathbb{Z}}$ with $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$, where each random variable X_t takes one of a finite number of *unordered* categories, is said to be a categorical process, and realizations $(x_t)_{t=1, \dots, T}$ are referred to as a categorical time series. To simplify notations, it is assumed that the range of $(X_t)_{\mathbb{N}}$ is coded as $\mathcal{V} = \{0, \dots, m\}$. This, however, does *not* imply that there is any natural order between the values of $\mathcal{V}$, except a lexicographic order. As a consequence, properties like dispersion and serial dependence have to be expressed by using “absolute measures” instead of “metric measures” (Blyth, 1959, p. 71), i.e., such measures can only rely on probabilities but not on the explicit values of the range. This problem was also considered by Weiß and GÖb (2008a) when defining weak forms of stationarity for categorical processes, while the usual definition of strict stationarity is “absolute” anyway: $(X_t)_{\mathbb{Z}}$ is said to be (*strictly*) *stationary* if the joint distribution of $(X_t, \dots, X_{t-k})$ is independent of t for all $k \in \mathbb{N}_0$. The lack of a natural order within $\mathcal{V}$ is also the main reason for the difficulties in visualizing categorical time series, see Weiß and GÖb (2008b) and Weiß (2008).

Categorical time series have received increasing interest during the last years, and multiple examples have been reported from diverse fields of practice, see Section 2 in Weiß (2009) for a survey. Among others, they occur in biological sequence analysis when considering sequences of the $m+1 = 4$ DNA bases or $m+1 = 20$ amino acids, or in part-of-speech tagging when each word of a text is assigned its part of speech (out of finitely many options), or in network monitoring when we observe time series of alarm messages signaling one out of finitely many error types.

Note. The abbreviations p_i , $\mathbf{p}$, $p_{ij}(k)$, $p_{ij|k}(k)$ and $s_k(\mathbf{p})$ for marginal and bivariate probabilities of a stationary process $(X_t)_{\mathbb{Z}}$, which are used in the sequel to simplify notations, are defined as in Weiß (2011a):

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- $p_i := P(X_t = i)$, $\mathbf{p} := (p_0, \dots, p_m)^\top$, and $s_k(\mathbf{p}) := \sum_{j=0}^m p_j^k$ for $k \in \mathbb{N}$;
- $p_{ij}(k) := P(X_t = i, X_{t-k} = j)$ and $p_{ij}(k) := P(X_t = i | X_{t-k} = j)$.

Weiß and GÖb (2008a) introduced the concept of (signed) serial dependence as the categorical counterpart to the (signed) autocorrelation in a real-valued process, and they proposed several (“absolute”) measures for serial dependence. To measure the degree of signed serial dependence, we use two types of Cohen’s κ in the sequel, namely

$$\kappa(k) = \frac{\sum_{j=0}^m p_{ij}(k) - s_2(\mathbf{p})}{1 - s_2(\mathbf{p})}, \quad \text{or} \quad \kappa^*(k) = \frac{1}{m} \cdot \left(\sum_{j=0}^m p_{ij}(k) - 1 \right). \quad (1)$$

The latter version of Cohen’s κ (referred to as the modified κ) is discussed by Weiß (2009, 2011a). To measure unsigned serial dependence, we consider Cramer’s v as well as Goodman and Kruskal’s τ ,

$$v(k) = \sqrt{\frac{1}{m} \cdot \sum_{i,j=0}^m \frac{(p_{ij}(k) - p_i p_j)^2}{p_i p_j}}, \quad \tau(k) = \sqrt{\sum_{i,j=0}^m \frac{(p_{ij}(k) - p_i p_j)^2}{p_j (1 - s_2(\mathbf{p}))}}. \quad (2)$$

It is easily seen that $\tau(k)$ corresponds to $\sqrt{A_v^{(\tau)}(k)}$ in Weiß and GÖb (2008a). Further measures of unsigned serial dependence are discussed by Dehnert et al. (2003), Weiß and GÖb (2008a) and Biswas and Song (2009). It is worth mentioning that Bagnato et al. (2012) recommend an autodependence measure being closely related to $v(k)$ in (2) also for application to cardinal time series. Furthermore, Wang and Li (2011) consider the autopersistence function, which is also “absolute” in the above sense, as a measure of serial dependence for the special case of a binary time series.

Measures of serial dependence are particularly important for identifying an appropriate model for a stationary categorical process and for estimating its model parameters. As an example, the true model order and the model parameters of an NDARMA model (Jacobs and Lewis, 1983) are characterized through appropriate measures of serial dependence. This model can be understood as a categorical counterpart to the conventional autoregressive moving average (ARMA) model: on the one hand, the NDARMA model is well suited for the case of an *unordered* range, and on the other hand, it shows some features being similar to those of a conventional ARMA model. In particular, the serial dependence structure exhibits some analogies to the autocorrelation structure of ARMA processes, see Section 2.1 for more details. Afterwards in Section 2.2, we show that an NDARMA process is φ -mixing, which is an important property if one tries to derive the asymptotic distribution of certain types of statistics. This is exemplified by Sections 3–6. First, we consider statistics being related to the marginal distribution of an NDARMA process: the Gini index as well as the entropy for measuring the dispersion in Section 3, and the Pearson statistic for goodness-of-fit testing in Section 4. Then, in Sections 5 and 6, we extend this work to statistics being related to bivariate distributions of an NDARMA process, by considering different types of empirical measure for serial dependence. Finally, we conclude in Section 7 and outline issues for future research.

2. NDARMA processes

2.1. Definition and basic properties

Jacobs and Lewis (1983) introduced the NDARMA model as a discrete counterpart to the cardinal ARMA model, based on preliminary research by Jacobs and Lewis (1978a,b,c). The NDARMA model was further analyzed by Dehnert et al. (2003), Weiß and GÖb (2008a), Biswas and Song (2009), Weiß (2009, 2011a,b), Song et al. (2013) and others. Weiß and GÖb (2008a) showed that Definition 2.1.1 is equivalent to the one given by Jacobs and Lewis (1983).

Definition 2.1.1 (NDARMA Model). Let $(X_t)_{\mathbb{Z}}$ and $(\epsilon_t)_{\mathbb{Z}}$ be categorical processes with range $\mathcal{V} = \{0, \dots, m\}$. Let $(\epsilon_t)_{\mathbb{Z}}$ be independent and identically distributed (i.i.d.) with marginal distribution $P(\epsilon_t = j) = p_j > 0$ for $j = 0, \dots, m$. Each ϵ_t is assumed to be independent of $(X_s)_{s < t}$.

Define the i.i.d. random vectors

$$\mathbf{D}_t = (\alpha_{t,1}, \dots, \alpha_{t,p}, \beta_{t,0}, \dots, \beta_{t,q}) \sim \text{MULT}(1; \phi_1, \dots, \phi_p, \varphi_0, \dots, \varphi_q)$$

for $t \in \mathbb{Z}$, with $\varphi_q > 0$ and with $\phi_p > 0$ if $p \geq 1$. Each $\mathbf{D}_t$ is independent of $(\epsilon_t)_{\mathbb{Z}}$ and of $(X_s)_{s < t}$.

The process $(X_t)_{\mathbb{Z}}$ is said to be an NDARMA(p, q) process if it follows the recursion

$$X_t = \alpha_{t,1} \cdot X_{t-1} + \dots + \alpha_{t,p} \cdot X_{t-p} + \beta_{t,0} \cdot \epsilon_t + \dots + \beta_{t,q} \cdot \epsilon_{t-q}. \quad (3)$$

In the case $q = 0$, the process is said to be a DAR(p) process. In the case $p = 0$, i.e., if $X_t = \beta_{t,0} \cdot \epsilon_t + \dots + \beta_{t,q} \cdot \epsilon_{t-q}$, it is referred to as a DMA(q) process.

So like in the case of the conventional ARMA model, the recursion (3) is based on p past observations and $q + 1$ innovations, but which are now incorporated in terms of a probabilistic mixture. Jacobs and Lewis (1983) showed that an NDARMA

process can be represented equivalently as a multivariate Markov chain, see Section 2.2 for details, and proved the existence of a stationary solution of the recursion (3). In the following, we shall always consider such *stationary* NDARMA(p, q) processes.

Note. A recent survey of essential properties of the NDARMA model can be found in Section 2 of Weiß (2011b). In the sequel, we only highlight some important facts but refer to this survey for more details.

The marginal distribution of the observations $(X_t)_{\mathbb{Z}}$ from a stationary NDARMA process is the same as that of the innovations $(\epsilon_t)_{\mathbb{Z}}$ (Jacobs and Lewis, 1983). Concerning the serial dependence structure, it is important to note that NDARMA processes only show positive serial dependence, i.e., $\kappa(k), \kappa^*(k) \geq 0$. Furthermore, the following equality holds (Weiß and GÖb, 2008a):

$$\kappa(k) = \kappa^*(k) = v(k) = \tau(k). \quad (4)$$

For this reason, we may simply use κ to express the serial dependence properties of an NDARMA process. This serial dependence structure, as already mentioned in Section 1, shows some analogies to the autocorrelation structure of conventional ARMA processes, in the sense that $\kappa(k)$ (or any other measure from (4)) is determined by a set of *Yule–Walker-type equations*:

$$\kappa(k) = \sum_{j=1}^p \phi_j \cdot \kappa(|k-j|) + \sum_{i=0}^{q-k} \varphi_{i+k} \cdot r(i) \quad \text{for } k \geq 1. \quad (5)$$

Then the pairwise conditional probabilities are obtained as (δ_{ij} is the Kronecker delta)

$$p_{ij}(k) = p_i \cdot (1 - \kappa(k)) + \delta_{ij} \cdot \kappa(k) = p_i + (\delta_{ij} - p_i) \cdot \kappa(k). \quad (6)$$

2.2. Markov chain representation and mixing properties

Let $(X_t)_{\mathbb{Z}}$ follow an NDARMA(p, q) model according to Definition 2.1.1. Then $(\mathbf{Z}_t)_{\mathbb{Z}}$ with

$$\mathbf{Z}_t := (X_t; X_{t-1}, \dots, X_{t-p+1}, \epsilon_t, \dots, \epsilon_{t-q+1})^\top \quad (7)$$

is a homogeneous Markov chain, see Jacobs and Lewis (1983), with transition probabilities given by

$$\begin{aligned} P(\mathbf{Z}_t = (i_0, \dots, i_{p-1}, j_0, \dots, j_{q-1}) | \mathbf{Z}_{t-1} = (i'_1, \dots, i'_p, j'_1, \dots, j'_q)) \\ = P(X_t = i_0 | X_{t-1} = i'_1, \dots, X_{t-p} = i'_p, \epsilon_t = j_0, \epsilon_{t-1} = j'_1, \dots, \epsilon_{t-q} = j'_q) \\ \cdot P(X_{t-1} = i_1 | X_{t-1} = i'_1) \cdots P(\epsilon_t = j_0) \cdot P(\epsilon_{t-1} = j_1 | \epsilon_{t-1} = j'_1) \cdots \\ = p_{j_0} \cdot \delta_{i_1 i'_1} \cdots \delta_{j_{q-1} j'_{q-1}} \cdot \left(\sum_{r=1}^p \delta_{i_0 i'_r} \phi_r + \delta_{i_0 j_0} \varphi_0 + \sum_{r=1}^q \delta_{i_0 j'_r} \varphi_r \right) \end{aligned} \quad (8)$$

according to recursion (3). So in the case of stationarity, the overall process distribution of $(\mathbf{Z}_t)_{\mathbb{Z}}$ is completely determined by the above transition probabilities (8) and the marginal distribution of $(\mathbf{Z}_t)_{\mathbb{Z}}$, i.e., the joint distribution of $(X_t; X_{t-1}, \dots, X_{t-p+1}, \epsilon_t, \dots, \epsilon_{t-q+1})$. The marginal distribution itself is determined from the transition probabilities (8) by solving an appropriate system of linear equations. Although this task can lead to tedious computations, we are finally always able to determine the overall distribution of $(X_t, \epsilon_t)_{\mathbb{Z}}$ and of $(X_t)_{\mathbb{Z}}$ for any NDARMA(p, q) process. Closed-form expressions for $p, q \leq 2$ are provided in Section 11.2.6 of Weiß (2009).

Theorem 2.2.1, see Appendix A.1 for the proof, uses the Markov chain representation $(\mathbf{Z}_t)_{\mathbb{Z}}$ of $(X_t)_{\mathbb{Z}}$ to establish diverse mixing properties for NDARMA processes.

Theorem 2.2.1 (Mixing Properties). *Let $(X_t)_{\mathbb{Z}}$ be an NDARMA(p, q) process according to Definition 2.1.1, and let $p + q \geq 1$. Then the process is ψ - and φ -mixing with exponentially decreasing weights $(f_n)_{\mathbb{N}}$, i.e., there exist $a > 0$ and $0 < \rho < 1$ such that $f_n = a \cdot \rho^n$. In particular, $\sum_{n \in \mathbb{N}} f_n^{1/2} < \infty$.*

An analogous result for DARMA($1, q + 1$) processes was derived by Jacobs and Lewis (1978b). Theorem 2.2.1 is particularly important for practice because of the following type of the central limit theorem, see p. 177 in Billingsley (1968).

If we have given a stationary φ -mixing process $(\mathbf{X}_t)_{\mathbb{Z}}$ of Borel random variables, where the sequence $(f_n)_{\mathbb{N}}$ satisfies $\sum_{n \in \mathbb{N}} f_n^{1/2} < \infty$, where $E[\mathbf{X}_t] = \mathbf{0}$ and $V[X_{t,i}] < \infty$ for all i , then $n^{-1/2} \cdot \sum_{k=1}^n \mathbf{X}_k$ is asymptotically distributed according to $N(\mathbf{0}, \Sigma)$, with $\Sigma = (\sigma_{ij})$ given by

$$\sigma_{ij} = E[X_{0,i} X_{0,j}] + \sum_{k=1}^{\infty} (E[X_{0,i} X_{k,j}] + E[X_{k,i} X_{0,j}]). \quad (9)$$

In the remaining Sections 3–6, we shall apply this result to derive asymptotic properties of several types of statistic being related to NDARMA(p, q) processes. These are based on the estimators

$$\hat{p}_i(T) := \frac{1}{T} \cdot \sum_{t=1}^T Y_{t,i} \quad \text{and} \quad \hat{p}_{ij}(k, T) := \frac{1}{T-k} \cdot \sum_{t=k+1}^T Y_{t,i} Y_{t-k,j} \quad (10)$$

for p_i and $p_{ij}(k)$, respectively, see (5) and (A8) in Weiß (2011a). Here, $\mathbf{Y}_t \in \{0, 1\}^{m+1}$ denotes the *binarization* of X_t , i.e., $Y_{t,i} := \delta_{i,X_t}$ for $i \in \mathcal{V}$.

Remark 2.2.2. When investigating the asymptotic properties of such statistics, it shall turn out that the result

$$c := 1 + 2 \cdot \sum_{k=1}^{\infty} \kappa(k) < \infty \quad (11)$$

is useful, see the proof in Appendix A.2. Particular instances of c , given a concrete NDARMA model, can be computed using (5). For instance, we obtain for a DAR(1) model that

$$c = 1 + 2 \cdot \sum_{k=1}^{\infty} \phi_1^k = 1 + \frac{2\phi_1}{1-\phi_1} = \frac{1+\phi_1}{1-\phi_1}, \quad (12)$$

and for a DMA(q) model that

$$\begin{aligned} c &= 1 + 2 \cdot \sum_{k=1}^q \sum_{i=0}^{q-k} \varphi_i \cdot \varphi_{i+k} = 1 + 2 \cdot \sum_{i=0}^{q-1} \sum_{j=i+1}^q \varphi_i \cdot \varphi_j \\ &= 1 + \left(\underbrace{\sum_{i,j=0}^q \varphi_i \varphi_j}_{=1} - \sum_{i=0}^q \varphi_i^2 \right) = 2 - \sum_{i=0}^q \varphi_i^2, \end{aligned} \quad (13)$$

also see Example 2.3 in Weiß (2011b). In the i.i.d. case, simply $c = 1$.

3. Measuring dispersion in NDARMA processes

As a first application of the theory derived in Section 2.2, we consider statistics being related to the marginal distribution of a stationary NDARMA process $(X_t)_{\mathbb{Z}}$, i.e., statistics being based on the estimators $\hat{p}_i(T)$ of p_i according to formula (10). The process $(\mathbf{X}_t)_{\mathbb{Z}}$ defined by $\mathbf{X}_t := \mathbf{Y}_t - \mathbf{p}$ has the marginal mean $\mathbf{0}$, and it satisfies the conditions for the asymptotic result (9), see Theorem 2.2.1. This implies that

$$\sqrt{T} \cdot (\hat{\mathbf{p}}(T) - \mathbf{p}) \xrightarrow{D} N(\mathbf{0}, c \cdot \Sigma) \quad \text{with } \sigma_{ij} = p_i \cdot (\delta_{ij} - p_j), \quad (14)$$

see the proof in Appendix B.1. This result, together with the delta theorem (Serfling, 1980, Section 3.3), allows us to derive asymptotic expressions for the distribution of statistics being related to the marginal distribution. We exemplify this in the following by considering the empirical Gini index and the empirical entropy (this section) as well as Pearson's χ^2 -statistic (Section 4).

3.1. Gini index

A popular approach for quantifying the dispersion of a categorical random variable is to use the Gini index $v_G(X) = \frac{m+1}{m} \cdot (1 - s_2(\mathbf{p}))$, also see Appendix A in Weiß and Göb (2008a). Given a segment $X_1, \dots, X_T$ with $T \in \mathbb{N}$ from a stationary NDARMA(p, q) process, it can be estimated by the *empirical Gini index*

$$\hat{v}_G(X) := \frac{m+1}{m} \cdot \frac{T}{T-1} \cdot (1 - s_2(\hat{\mathbf{p}}(T))), \quad (15)$$

see Weiß (2011a). In the i.i.d. case, this is an exactly unbiased estimator of $v_G(X)$, and Weiß (2011a) also proved the asymptotic normality for this case. We now analyze the asymptotic distribution of $\hat{v}_G(X)$ for the more general NDARMA case.

Theorem 3.1.1 (Gini Index). Let $X_1, \dots, X_T$ with $T \in \mathbb{N}$ be a segment from a stationary NDARMA(p, q) process according to Definition 2.1.1 with $p + q \geq 1$.

Table 1

Empirical mean (“M”) and standard deviation (“SD”) of simulated empirical Gini indices, plus related asymptotic standard deviation (“SD_a”). “M_a” refers to the approximation $\frac{T-c}{T-1} \cdot \nu_G(X)$ according to formula (17).

Marginal:		$\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$				$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$			
Gini index:		$\nu_G = 0.98$				$\nu_G = 0.6333 \dots$			
Model	T	M	M _a	SD	SD _a	M	M _a	SD	SD _a
i.i.d. ($c = 1$)	100	0.980	0.980	0.019	0.017	0.633	0.633	0.072	0.071
	200	0.980	0.980	0.013	0.012	0.633	0.633	0.050	0.050
	500	0.980	0.980	0.008	0.008	0.633	0.633	0.032	0.032
	1000	0.980	0.980	0.006	0.006	0.633	0.633	0.023	0.023
DMA(1), $\varphi_1 = 0.25$ ($c = 1.375$)	100	0.976	0.976	0.023	0.020	0.630	0.631	0.083	0.083
	200	0.978	0.978	0.016	0.014	0.632	0.632	0.059	0.059
	500	0.979	0.979	0.010	0.009	0.633	0.633	0.038	0.037
	1000	0.980	0.980	0.007	0.006	0.633	0.633	0.026	0.026
DAR(1), $\phi_1 = 0.4$ ($c = 2.333$)	100	0.967	0.967	0.032	0.026	0.627	0.625	0.109	0.108
	200	0.973	0.973	0.021	0.019	0.628	0.629	0.076	0.077
	500	0.977	0.977	0.013	0.012	0.632	0.632	0.049	0.049
	1000	0.979	0.979	0.009	0.008	0.633	0.632	0.035	0.034
DAR(1), $\phi_1 = 0.8$ ($c = 9$)	100	0.906	0.901	0.081	0.052	0.587	0.582	0.193	0.212
	200	0.942	0.941	0.050	0.037	0.609	0.608	0.143	0.151
	500	0.965	0.964	0.027	0.023	0.623	0.623	0.093	0.096
	1000	0.972	0.972	0.018	0.017	0.629	0.628	0.067	0.068

Then the distribution of $1 - s_2(\hat{\mathbf{p}}(T))$ is asymptotically approximated by a normal distribution with mean $1 - s_2(\mathbf{p})$ and variance $\frac{4c}{T} \cdot (s_3(\mathbf{p}) - s_2^2(\mathbf{p}))$.

The proof of Theorem 3.1.1 is provided by Appendix B.2. In the case of i.i.d. data (i.e., if $c = 1$), the asymptotic variance given in Theorem 3.1.1 reduces to the one given in Lemma 2.1 of Weiß (2011a).

3.2. Entropy

An alternative measure of dispersion is the entropy $\nu_E(X) = -\frac{1}{\ln(m+1)} \sum_{j=0}^m p_j \ln p_j$, also see Appendix A in Weiß and Göb (2008a). It can be estimated by

$$\hat{\nu}_E(X) := -\frac{1}{\ln(m+1)} \sum_{j=0}^m \hat{p}_j(T) \ln \hat{p}_j(T), \quad (16)$$

where we use the convention $0 \cdot \ln 0 := 0$ for both $\nu_E(X)$ and $\hat{\nu}_E(X)$. We now analyze the asymptotic behavior of $\hat{\nu}_E(X)$.

Theorem 3.2.1 (Entropy). Let $X_1, \dots, X_T$ with $T \in \mathbb{N}$ be a segment from a stationary NDARMA(p, q) process according to Definition 2.1.1 with $p + q \geq 1$.

Then the distribution of $-\sum_{j=0}^m \hat{p}_j(T) \ln \hat{p}_j(T)$ is asymptotically approximated by a normal distribution with mean $-\sum_{j=0}^m p_j \ln p_j$ and variance

$$\frac{c}{T} \cdot \left(\sum_{i=0}^m p_i \cdot (\ln p_i)^2 - \left(\sum_{i=0}^m p_i \cdot \ln p_i \right)^2 \right).$$

The proof of Theorem 3.2.1 is provided by Appendix B.3. In the case of i.i.d. data (i.e., if $c = 1$), the asymptotic variance given in Theorem 3.2.1 coincides with the one given on p. 75 in Blyth (1959).

3.3. Finite-sample properties

For applications in practice, it is important to know if the asymptotic results 3.1.1 and 3.2.1 can serve as a reasonable approximation to the true distribution of $\hat{\nu}_G(X)$ and $\hat{\nu}_E(X)$ given a time series of finite length T . Therefore, we simulated different types of NDARMA processes (always 10,000 replications per situation, same marginal distributions as in Weiß, 2011a) and computed these measures from segments of lengths $T = 100, 200, 500$ or 1000. To save space, only selected results are presented in the sequel, but further results can be obtained from the author upon request.

Tables 1 and 2 show the empirically obtained values for mean and standard deviation of the empirical Gini index and entropy for the cases of independence, DMA(1) dependence and DAR(1) dependence. The degree of serial dependence increases from block to block in Tables 1 and 2, also see Example 2.3 in Weiß (2011b). While the Gini index is exactly

Table 2

Empirical mean (“M”) and standard deviation (“SD”) of simulated empirical entropies, plus related asymptotic standard deviation (“SD_a”).

Marginal:		$\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$			$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$		
Entropy:		$v_E = 0.9794 \dots$			$v_E = 0.6595 \dots$		
Model	T	M	SD	SD _a	M	SD	SD _a
i.i.d. ($c = 1$)	100	0.969	0.019	0.017	0.648	0.064	0.063
	200	0.974	0.013	0.012	0.653	0.045	0.045
	500	0.977	0.008	0.008	0.657	0.029	0.028
	1000	0.978	0.006	0.006	0.658	0.020	0.020
DMA(1), $\phi_1 = 0.25$ ($c = 1.375$)	100	0.964	0.024	0.020	0.644	0.074	0.074
	200	0.972	0.016	0.014	0.652	0.053	0.053
	500	0.977	0.010	0.009	0.656	0.034	0.033
	1000	0.978	0.007	0.006	0.658	0.023	0.024
DAR(1), $\phi_1 = 0.4$ ($c = 2.333$)	100	0.954	0.033	0.027	0.636	0.098	0.097
	200	0.967	0.022	0.019	0.646	0.068	0.068
	500	0.974	0.013	0.012	0.655	0.043	0.043
	1000	0.977	0.009	0.008	0.657	0.031	0.031
DAR(1), $\phi_1 = 0.8$ ($c = 9$)	100	0.883	0.087	0.052	0.569	0.176	0.190
	200	0.931	0.052	0.037	0.611	0.129	0.134
	500	0.960	0.027	0.023	0.640	0.084	0.085
	1000	0.970	0.018	0.017	0.650	0.060	0.060

Table 3

Empirical skewness and excess of simulated empirical Gini indices.

Marginal:		$\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$				$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$		
Model	i.i.d.	DMA(1)	DAR(1)	DAR(1)	i.i.d.	DMA(1)	DAR(1)	DAR(1)
T		$\varphi_1 = 0.25$	$\phi_1 = 0.4$	$\phi_1 = 0.8$		$\varphi_1 = 0.25$	$\phi_1 = 0.4$	$\phi_1 = 0.8$
100	−1.16	−1.22	−1.49	−1.61	−0.23	−0.23	−0.29	−0.44
200	−0.86	−1.00	−1.21	−1.63	−0.11	−0.18	−0.17	−0.35
500	−0.56	−0.77	−0.86	−1.32	−0.12	−0.13	−0.14	−0.23
1000	−0.46	−0.48	−0.65	−1.16	−0.02	−0.06	−0.11	−0.16
100	2.06	2.15	3.28	3.94	−0.05	−0.02	−0.02	−0.33
200	1.08	1.38	2.13	4.31	0.02	−0.02	−0.01	−0.14
500	0.40	1.00	1.00	2.37	−0.02	0.10	−0.04	−0.08
1000	0.29	0.28	0.61	2.05	0.01	−0.08	0.00	−0.02

Table 4

Empirical skewness and excess of simulated empirical entropies.

Marginal:		$\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$				$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$		
Model	i.i.d.	DMA(1)	DAR(1)	DAR(1)	i.i.d.	DMA(1)	DAR(1)	DAR(1)
T		$\varphi_1 = 0.25$	$\phi_1 = 0.4$	$\phi_1 = 0.8$		$\varphi_1 = 0.25$	$\phi_1 = 0.4$	$\phi_1 = 0.8$
100	−1.07	−1.15	−1.34	−1.27	−0.23	−0.24	−0.28	−0.33
200	−0.78	−0.91	−1.09	−1.35	−0.10	−0.18	−0.16	−0.29
500	−0.52	−0.72	−0.80	−1.19	−0.13	−0.13	−0.12	−0.22
1000	−0.42	−0.45	−0.60	−1.07	−0.03	−0.08	−0.11	−0.15
100	1.69	1.91	2.55	2.07	0.00	0.01	0.05	−0.21
200	0.80	1.03	1.65	2.54	0.02	0.02	0.07	−0.07
500	0.29	0.85	0.86	1.89	−0.02	0.14	−0.03	−0.01
1000	0.23	0.21	0.51	1.75	0.01	−0.07	−0.04	−0.02

unbiased (by construction) in the i.i.d. case, it shows increasing bias for increasing dependence, with a stronger effect for the highly-dispersed situation $\mathbf{p}_1$. Especially for $\phi_1 = 0.8$ and $T \leq 200$, this bias is not negligible anymore (up to one standard deviation). The bias shown by the entropy is even stronger. It becomes visible already in the i.i.d. situation for $T \leq 200$, and for $\phi_1 = 0.8$, there is still a substantial bias even for $T = 1000$.

A similar behavior, but less pronounced, is observed for the empirical standard deviations (compared to the asymptotic ones), so we only highlight an additional feature: while the Gini index shows slightly less standard deviation than the entropy for the highly-dispersed situation $\mathbf{p}_1$, the entropy shows clearly less standard deviation in the situation $\mathbf{p}_2$ with only moderate dispersion. Finally, the goodness of the normal approximation was investigated. Using the 10,000 replications per situation, empirical skewness and excess were computed (according to the formulas for G_1 , G_2 in Joanes and Gill, 1998, p. 184), see Tables 3 and 4, and quantile plots were generated (two of them are shown in Fig. 1 for illustrative purposes,

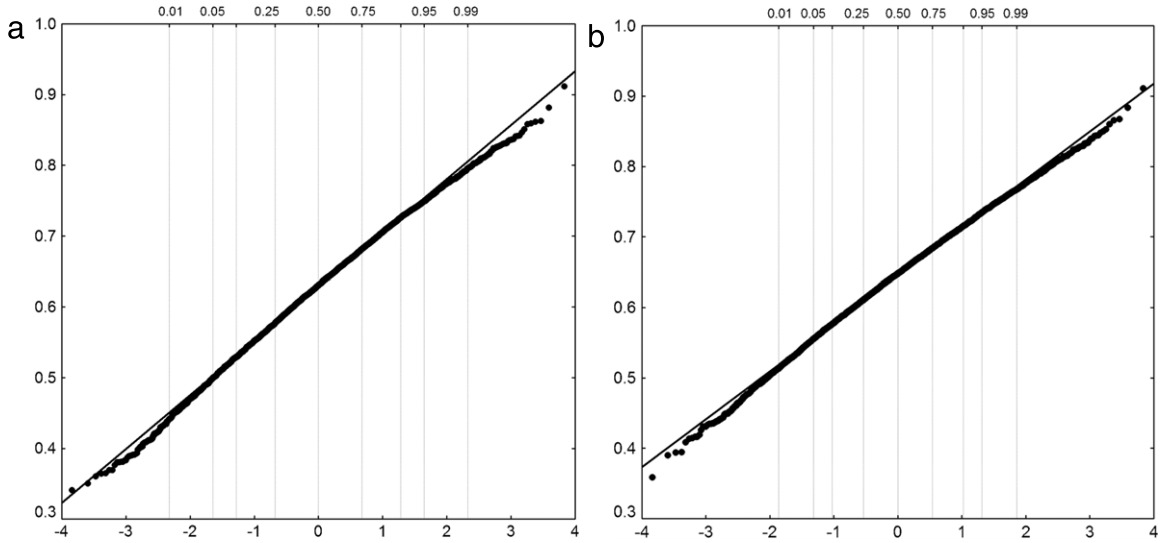


Fig. 1. Quantile plot of simulated (a) Gini indices and (b) entropies ($T = 200$) for DAR(1) model with $\phi_1 = 0.4$ and marginal distribution $\mathbf{p}_2$.

the remaining plots are available upon request). It becomes clear that the normal approximation is clearly worse for the highly-dispersed situation $\mathbf{p}_1$, but which is reasonable, since most of the probability mass for Gini index and entropy is concentrated close to the upper bound 1 of their range. For the moderately-dispersed situation $\mathbf{p}_2$, the normal approximation works reasonably well at least for $T \geq 200$, also see Fig. 1. Overall, we recognize that the normal approximation for the entropy is slightly better than for the Gini index.

In a nutshell, especially the large bias of both dispersion measures for strong degrees of serial dependence and $T \leq 200$ is problematic. If it would be possible to find a better approximation of $E[\hat{v}_G(X)]$, $E[\hat{v}_E(X)]$ than simply $v_G(X)$, $v_E(X)$, this approximation could, in turn, be applied to correct the bias. In fact, since $\hat{v}_G(X)$ is just a quadratic polynomial of the $\hat{p}_i(T)$, we can even determine $E[\hat{v}_G(X)]$ exactly, see Appendix B.4:

$$E[\hat{v}_G(X)] = v_G(X) \cdot \frac{T}{T-1} \cdot \left(1 - \frac{1}{T} \cdot \left(1 + 2 \sum_{k=1}^{T-1} \left(1 - \frac{k}{T} \right) \cdot \kappa(k) \right) \right) \geq v_G(X) \cdot \frac{T-c}{T-1}. \quad (17)$$

Since for any NDARMA model, we can determine $\kappa(k)$ according to (5), we can also determine $E[\hat{v}_G(X)]$ exactly. However, formula (17) also implies that $\frac{T-c}{T-1} \cdot v_G(X)$ might be used as a simple approximation for the mean (which even contains the i.i.d. case exactly). As we can see from Table 1, this approximation is indeed very close to the empirical mean such that its use is clearly recommended. An analogous approximation for the entropy, however, is more problematic, since simple but accurate polynomial approximations of $p \ln p$ are not available. But it could at least be verified from Table 2 that the bias of the entropy is reduced by considering $v_E(X) \cdot \frac{T-c}{T-1}$ as a mean approximation, i.e., by using the same correction as for the Gini index.

3.4. Real-data example

A common approach for the stochastic modeling of DNA sequences is to use appropriate types of Markov models (Usotskaya and Ryabko, 2009). Among others, the use of the DAR(p) model (as a sparsely parametrized pth order Markov model) has been recommended for this purpose (Dehnert et al., 2003). We consider a DNA sequence corresponding to the Bovine leukemia virus (length $T = 8419$), which was already analyzed by Weiß and Göb (2008a). Its range is of size $m + 1 = 4$ and consists of the DNA bases ‘g’, ‘a’, ‘t’ and ‘c’ (coded by 0, 1, 2 and 3, respectively). The corresponding marginal distribution $\hat{\mathbf{p}}$ is estimated by the relative frequencies $\hat{p}_0 = 0.210$, $\hat{p}_1 = 0.220$, $\hat{p}_2 = 0.239$ and $\hat{p}_3 = 0.331$ (Weiß and Göb, 2008a, Example 6.1). Looking at the first few values (Weiß and Göb, 2008a, Example 1.1) or at a plot of the data (Weiß, 2008, Figure 1), it becomes clear that this time series is characterized by the occurrence of runs of symbols, which, in turn, indicates positive dependence. In fact, Weiß and Göb (2008a) showed that the data can be modeled quite well by a DAR(1) or DAR(2) model. For illustrative purposes, we shall now consider a DAR(1) model for the data, with the ML-estimated parameter $\hat{\phi}_1 \approx 0.081$. As a consequence, $\hat{c} \approx 1.18$ according to (12).

Using (15) and (16) to estimate the Gini index and entropy, respectively, we obtain the values 0.988 and 0.987 (high dispersion). Using Theorems 3.1.1 and 3.2.1, both standard errors are approximated by ≈ 0.0016 . So the empirically observed dispersion is not only significantly different from the dispersion of a uniform distribution ($v = 1$), but also from that of the distribution $\mathbf{p}_1$ discussed in Section 3.3.

Table 5

Empirical rejection rates (“r.r.”) for Pearson statistics with regard to the 95%-quantile of the χ^2_3 -distribution, 7.8147....

Marginal:		$\mathbf{p}_1$	$\mathbf{p}_2$	Marginal:		$\mathbf{p}_1$	$\mathbf{p}_2$
Model	T	r.r.	r.r.	Model	T	r.r.	r.r.
i.i.d. ($c = 1$)	100	0.049	0.048	DAR(1), $\phi_1 = 0.4$ ($c = 2.333$)	100	0.050	0.053
	200	0.048	0.051		200	0.048	0.051
	500	0.047	0.054		500	0.048	0.050
	1000	0.050	0.051		1000	0.051	0.054
DMA(1), $\phi_1 = 0.25$ ($c = 1.375$)	100	0.049	0.050	DAR(1), $\phi_1 = 0.8$ ($c = 9$)	100	0.042	0.063
	200	0.052	0.050		200	0.042	0.058
	500	0.052	0.052		500	0.042	0.052
	1000	0.052	0.049		1000	0.048	0.053

4. Pearson's χ^2 -statistic for NDARMA processes

To test a hypothesis concerning the marginal distribution of random variables, one of the most popular approaches is to apply Pearson's χ^2 -statistic, which is known to asymptotically follow a χ^2 -distribution in the i.i.d. case. We extend this result for general stationary NDARMA processes by applying formula (14), also see Theorem 11.2.5.3 in Weiß (2009).

Theorem 4.1 (Pearson's χ^2 -Statistic). Let $X_1, \dots, X_T$ with $T \in \mathbb{N}$ be a segment from a stationary NDARMA(p, q) process according to Definition 2.1.1 with $p + q \geq 1$. We use the notations from formulas (10) and (11). Then Pearson's χ^2 -statistic X^2 satisfies

$$X^2/c = \frac{T}{c} \cdot \sum_{i=0}^m \frac{(\hat{p}_i(T) - p_i)^2}{p_i} \xrightarrow{D} \chi_m^2.$$

The proof is done as in Example 3.5.B of Serfling (1980) by using formula (14). An analogous result for DARMA($1, q + 1$) processes was derived by Jacobs and Lewis (1978b). Theorem 4.1 also includes the well-known i.i.d.-result, since then $c = 1$.

To analyze the finite-sample performance of the approximation in Theorem 4.1, the simulation study described in Section 3 before was extended to the Pearson statistic. In view of the main application of this statistic, we do not show values for mean and standard deviation this time (but these and further statistics are available upon request). We now fix the 95%-quantile of the respective χ^2 -distribution ($\approx$ critical value for a goodness-of-fit test) and determine the rate of simulation runs, where the value of the Pearson statistic falls above this quantile (empirical rejection rate). The results of Table 5 show that the nominal level of 5% rejections is usually met quite well, only in the case of the strong dependence (DAR(1) model with $\phi_1 = 0.8$), we observe clear deviations for $T \leq 500$. While the rejection rates are relatively robust against serial dependence, the goodness of the χ^2 -approximation as a whole suffers from the presence of serial dependence: the quantile plots for the simulated Pearson statistics according to the DAR(1) models (not shown but available upon request) exhibit strong deviations from the χ^2 -assumption especially for the moderately-dispersed situation $\mathbf{p}_2$, i.e., quite the opposite behavior than it is observed for the statistics in Section 3.

Example 4.2 (Bovine Data). For the Bovine sequence discussed in Section 3.4, we already noted that it shows a dispersion behavior being significantly different from that of the distribution $\mathbf{p}_1$. Nevertheless, we now use $\mathbf{p}_1$ as the null for computing $X^2/\hat{c}$, leading to a value of about 44.9. Since this value is larger than the 95%-quantile of the χ^2_3 -distribution, 7.8147, we again reject the hypothesis that the Bovine data has the true marginal distribution $\mathbf{p}_1$.

5. Empirical measures of signed serial dependence

The theory derived in Section 2.2 can also be applied to statistics being related to bivariate distributions of a stationary NDARMA process $(X_t)_{\mathbb{Z}}$, i.e., to statistics, which are based on the estimators $\hat{p}_i(T)$ of p_i and $\hat{p}_{ij}(k, T)$ of $p_{ij}(k)$, see formula (10). The general idea of how to derive the asymptotics of such statistics is the same as in Section 3 before, but we have to consider a process $(\mathbf{X}_t)_{\mathbb{Z}}$, where each $\mathbf{X}_t$ is an appropriate function of $\mathbf{Y}_t$ and $\mathbf{Y}_{t-k}$. Since $\mathbf{X}_t$ is still a function of finitely many variables from an NDARMA process, Theorem 2.2.1 implies that also $(\mathbf{X}_t)_{\mathbb{Z}}$ is φ -mixing with geometrically decreasing weights. This shall allow us to again apply the asymptotic result (9).

Remark 5.1 (Abbreviations). In the sequel, we adapt the abbreviation $p_{ij}(k)$ for the bivariate probabilities also to the case of trivariate and quadrivariate probabilities:

$$\begin{aligned} p_{abc}(k, l) &:= P(X_t = a, X_{t-k} = b, X_{t-k-l} = c), \\ p_{abcd}(k, l, m) &:= P(X_t = a, X_{t-k} = b, X_{t-k-l} = c, X_{t-k-l-m} = d). \end{aligned}$$

For estimating the measures of signed serial dependence from (1) (with a fixed $k \geq 1$), we do not require the estimators $\hat{p}_{ij}(k, T)$ with $i \neq j$, i.e., it suffices to define the process $(\mathbf{X}_t)_{\mathbb{Z}}$ by

$$\begin{aligned} \mathbf{X}_t &:= \mathbf{X}_t(k) := (X_{t;0}, \dots, X_{t;m}, X_{t;00}(k), \dots, X_{t;mm}(k))^{\top} \in \mathbb{R}^{2(m+1)} \\ \text{with } X_{t;i} &:= Y_{t,i} - p_i, \quad X_{t;ii}(k) := Y_{t,i}Y_{t-k,i} - p_{ii}(k). \end{aligned} \quad (18)$$

We obtain that $\sqrt{T} \cdot \left(\frac{1}{T} \sum_{t=1}^T \mathbf{X}_t \right)$ is asymptotically distributed according to $N(\mathbf{0}, \Sigma)$, where the covariance matrix Σ is a block matrix

$$\begin{aligned} \Sigma &= \left(\begin{array}{c|c} \sigma_{ij} & \sigma_{i;bb} \\ \hline \sigma_{aa;j} & \sigma_{aa;bb} \end{array} \right) \quad \text{with } \sigma_{ij} = p_j \cdot (\delta_{ij} - p_i) \cdot c, \\ \sigma_{i;bb} &= 2(\delta_{ib} - p_i) \cdot p_{bb}(k) + \sum_{n=1}^{k-1} (p_{bib}(n, k-n) - p_i p_{bb}(k)) \\ &\quad + \sum_{n=k+1}^{\infty} (p_{bbi}(k, n-k) - p_i p_{bb}(k)) + \sum_{n=1}^{\infty} (p_{ibb}(n, k) - p_i p_{bb}(k)), \\ \sigma_{aa;bb} &= (\delta_{ab} - p_{bb}(k)) \cdot p_{aa}(k) + \sum_{n=1}^{k-1} (p_{baba}(n, k-n, n) + p_{abab}(n, k-n, n) - 2p_{aa}(k)p_{bb}(k)) \\ &\quad + 2(\delta_{ab} \cdot p_{abb}(k, k) - p_{aa}(k)p_{bb}(k)) + \sum_{n=k+1}^{\infty} (p_{bbaa}(k, n-k, k) + p_{aabb}(k, n-k, k) - 2p_{aa}(k)p_{bb}(k)). \end{aligned} \quad (19)$$

If the underlying process $(X_t)_{\mathbb{Z}}$ is even i.i.d., then the entries of Σ simplify to

$$\sigma_{i;bb} = 2(\delta_{ib} - p_i)p_b^2, \quad \sigma_{aa;bb} = \delta_{ab} \cdot p_a^2 \cdot (1 + 2p_a) - 3p_a^2 p_b^2. \quad (20)$$

The proof of formulas (19) and (20) is postponed to Section 6, where we consider a more general result.

5.1. Cohen's κ

Weiß (2011a) proposed the bias-corrected estimator

$$\hat{\kappa}(k) := 1 + \frac{1}{T} - \frac{1 - \sum_{j=0}^m \hat{p}_{jj}(k, T)}{1 - s_2(\hat{\mathbf{p}}(T))} \quad (21)$$

of Cohen's $\kappa(k)$ according to (1), and he analyzed its asymptotic behavior in the i.i.d. case. We now extend these investigations to the case of an NDARMA process.

Theorem 5.1.1 (Cohen's κ). *Let $X_1, \dots, X_T$ be a segment from a stationary NDARMA(p, q) process according to Definition 2.1.1 with $p + q \geq 1$. Let $k \in \mathbb{N}$.*

Then the distribution of the empirical Cohen's κ of (21), $\hat{\kappa}(k)$, is asymptotically approximated by a normal distribution with mean $\kappa(k)$ and variance

$$\frac{1}{T} \left(4c \cdot \frac{\left(1 - \sum_{j=0}^m p_{jj}(k) \right)^2}{(1 - s_2(\mathbf{p}))^4} \cdot (s_3(\mathbf{p}) - s_2^2(\mathbf{p})) - 4 \cdot \frac{1 - \sum_{j=0}^m p_{jj}(k)}{(1 - s_2(\mathbf{p}))^3} \cdot \sum_{i,b=0}^m p_i \cdot \sigma_{i;bb} + \frac{1}{(1 - s_2(\mathbf{p}))^2} \cdot \sum_{a,b=0}^m \sigma_{aa;bb} \right).$$

The proof of Theorem 5.1.1 is provided by Appendix C.1. As exemplified in formula (C.1) in Appendix C.1, the expression for the asymptotic variance in Theorem 5.1.1 indeed simplifies in the i.i.d. case to the expression known from Theorem 2.3 in Weiß (2011a).

5.2. Modified κ

Weiß (2011a) proposed the bias-corrected estimator

$$\hat{\kappa}^*(k) := \frac{1}{T} + \frac{1}{m} \cdot \left(\sum_{j=0}^m \frac{\hat{p}_{jj}(k, T)}{\hat{p}_j(T)} - 1 \right) \quad (22)$$

Table 6

Number of “invalid” DAR(1) time series while simulating 10,000 valid time series.

Marginal: $T \setminus \phi_1$	$\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$				$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$			
	0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8
100	0	0	3	360	173	499	1692	8555
200	0	0	0	6	1	17	183	1660
500	0	0	0	0	0	0	0	73
1000	0	0	0	0	0	0	0	1

of the measure $\kappa^*(k)$ according to (1), and he analyzed its asymptotic behavior in the i.i.d. case. We now extend these investigations to the case of an NDARMA process.

Theorem 5.2.1 (Modified κ). Let $X_1, \dots, X_T$ be a segment from a stationary NDARMA(p, q) process according to Definition 2.1.1 with $p + q \geq 1$. Let $k \in \mathbb{N}$.

Then the distribution of the empirical modified κ of (22), $\hat{\kappa}^*(k)$, is asymptotically approximated by a normal distribution with mean $\kappa(k)$ and variance

$$\frac{1}{m^2 \cdot T} \left(c \cdot \sum_{i=0}^m \frac{p_{ii}(k)^2}{p_i^3} - c \cdot \left(\sum_{i=0}^m p_{ii}(k) \right)^2 - 2 \cdot \sum_{i,b=0}^m \frac{p_{ii}(k)}{p_i^2 p_b} \cdot \sigma_{i,bb} + \sum_{a,b=0}^m \frac{\sigma_{aa,bb}}{p_a p_b} \right).$$

The proof of Theorem 5.2.1 is provided by Appendix C.1. As exemplified in formula (C.2) in Appendix C.1, the expression for the asymptotic variance in Theorem 5.2.1 indeed simplifies in the i.i.d. case to the expression known from Theorem 2.4 in Weiß (2011a).

5.3. Parameter estimation for DAR(1) processes

In the case of the DAR(1) model, we have $\kappa(1) = \phi_1$ according to (5), which, in turn, implies that $\hat{\kappa}(1)$ and $\hat{\kappa}^*(1)$ can serve as estimators of the dependence parameter ϕ_1 . As a first application of Theorems 5.1.1 and 5.2.1, we now derive the asymptotic distribution of these estimators, see the proof in Appendix C.2.

Corollary 5.3.1 (DAR(1) Model: Estimator for ϕ_1). Let $X_1, \dots, X_T$ be a segment from a stationary DAR(1) process according to Definition 2.1.1.

(a) The distribution of the estimator $\hat{\phi}_{1,C} := \hat{\kappa}(1)$ of ϕ_1 is asymptotically approximated by a normal distribution with mean ϕ_1 and variance

$$\frac{1}{T} \left(2 \cdot (2\phi_1 - 1) \cdot (1 - \phi_1) \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} + \frac{(1 - \phi_1) \cdot s_2(\mathbf{p})}{1 - s_2(\mathbf{p})} + \phi_1(1 - \phi_1) \right).$$

(b) The distribution of the estimator $\hat{\phi}_{1,M} := \hat{\kappa}^*(1)$ of ϕ_1 is asymptotically approximated by a normal distribution with mean ϕ_1 and variance

$$\frac{1}{m^2 \cdot T} \left(m \cdot (1 - \phi_1)^2 + \phi_1(1 - \phi_1) \cdot \sum_{i=0}^m \frac{1 - p_i}{p_i} \right).$$

The finite-sample performance of these estimators was analyzed in a simulation study like in Section 3, but now only DAR(1) models were simulated with dependence parameters $\phi_1 = 0.2, 0.4, 0.6, 0.8$. Before reporting selected results with regard to stochastic properties of the estimators $\hat{\phi}_{1,C}, \hat{\phi}_{1,M}$ (further results are available upon request), a computational issue has to be mentioned, which was negligible in the i.i.d. situation considered by Weiß (2011a), but which becomes more and more important for increasing degree of serial dependence. According to Theorem 2.3 in Weiß (2011a), the measure $\hat{\kappa}$ and hence also $\hat{\phi}_{1,C}$ can always be computed except for samples where $s_2(\hat{\mathbf{p}}(T)) = 1$, i.e., except for constant samples (which will rarely happen in practice). $\hat{\kappa}^*$ and $\hat{\phi}_{1,M}$, however, require all the $\hat{p}_j(T)$ to be truly positive, otherwise we are concerned with the problem of division by 0. To be able to compare $\hat{\phi}_{1,C}$ and $\hat{\phi}_{1,M}$ with each other, we declared a simulated time series as being “invalid” if $\min_j \{\hat{p}_j(T)\} = 0$. The numbers of such invalid time series being generated while simulating 10,000 valid time series are reported in Table 6. These numbers obviously increase for decreasing T , for decreasing dispersion and for increasing ϕ_1 . In many of the tabulated situations, we would often have the problem in practice that $\hat{\phi}_{1,M}$ cannot be computed at all, a first evidence voting against the use of the modified κ for the purpose of parameter estimation.

Next, we empirically analyze stochastic properties of the estimators $\hat{\phi}_{1,C}, \hat{\phi}_{1,M}$ among 10,000 valid time series per situation. Tables 7 and 8 show the empirically obtained values for mean and standard deviation of both estimators as well as the related asymptotic standard deviations. If we consider the bias of both estimators, we recognize (1) always a negative

Table 7

Empirical mean (“M”) and standard deviation (“SD”) of simulated estimator $\hat{\phi}_{1,C}$, plus related asymptotic standard deviation (“SD_a”).

Marginal:		$\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$			$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$		
ϕ_1	T	M	SD	SD _a	M	SD	SD _a
0.2	100	0.200	0.067	0.066	0.194	0.087	0.086
	200	0.199	0.047	0.047	0.197	0.061	0.061
	500	0.199	0.030	0.030	0.199	0.039	0.038
	1000	0.200	0.021	0.021	0.200	0.027	0.027
0.4	100	0.396	0.068	0.067	0.390	0.093	0.091
	200	0.398	0.048	0.048	0.395	0.065	0.064
	500	0.399	0.030	0.030	0.398	0.041	0.041
	1000	0.399	0.021	0.021	0.399	0.029	0.029
0.6	100	0.593	0.063	0.062	0.582	0.091	0.086
	200	0.597	0.044	0.044	0.592	0.063	0.061
	500	0.599	0.028	0.028	0.596	0.039	0.038
	1000	0.599	0.020	0.020	0.598	0.027	0.027
0.8	100	0.790	0.051	0.048	0.784	0.078	0.068
	200	0.796	0.035	0.034	0.789	0.054	0.048
	500	0.798	0.022	0.022	0.796	0.032	0.030
	1000	0.799	0.015	0.015	0.798	0.022	0.021

Table 8

Empirical mean (“M”) and standard deviation (“SD”) of simulated estimator $\hat{\phi}_{1,M}$, plus related asymptotic standard deviation (“SD_a”).

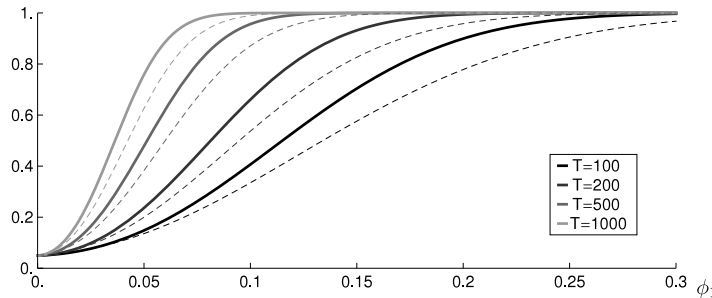
Marginal:		$\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$			$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$		
ϕ_1	T	M	SD	SD _a	M	SD	SD _a
0.2	100	0.195	0.067	0.066	0.177	0.088	0.091
	200	0.196	0.047	0.047	0.189	0.064	0.064
	500	0.198	0.030	0.030	0.196	0.041	0.040
	1000	0.199	0.021	0.021	0.198	0.029	0.029
0.4	100	0.386	0.070	0.068	0.352	0.107	0.101
	200	0.393	0.049	0.048	0.377	0.077	0.072
	500	0.397	0.030	0.030	0.392	0.047	0.045
	1000	0.399	0.022	0.022	0.395	0.033	0.032
0.6	100	0.574	0.070	0.063	0.519	0.122	0.098
	200	0.589	0.047	0.045	0.559	0.088	0.069
	500	0.596	0.029	0.028	0.585	0.049	0.044
	1000	0.598	0.020	0.020	0.593	0.032	0.031
0.8	100	0.752	0.078	0.049	0.718	0.135	0.079
	200	0.782	0.043	0.035	0.734	0.101	0.056
	500	0.794	0.023	0.022	0.776	0.052	0.035
	1000	0.797	0.016	0.016	0.790	0.030	0.025

bias, and (2) the same tendency as before, i.e., the bias becomes worse for decreasing T , for decreasing dispersion and for increasing ϕ_1 . But, while these effects are usually negligible for the estimator $\hat{\phi}_{1,C}$ (with moderate biases for $\phi_1 \geq 0.6$ and $T \leq 200$), we often observe a rather severe bias for $\hat{\phi}_{1,M}$ especially in the situation $\mathbf{p}_2$ with low dispersion. An analogous conclusion holds if we compare the empirical standard deviation with the asymptotic one. Only for the estimator $\hat{\phi}_{1,C}$, we can recommend the asymptotic standard deviation as a reasonable approximation to the true standard deviation in any of the considered situations. This drifting apart between $\hat{\phi}_{1,C}$ and $\hat{\phi}_{1,M}$ also becomes manifest in their empirical correlation, see Table 9. Only in view of the goodness of the normal approximation (in terms of quantile plots, skewness and excess; corresponding results are available upon request), both estimators behave quite similar and show clear deviations at least for $\mathbf{p}_2$ and $\phi_1 = 0.8$. So in a nutshell, while the modified κ might be attractive for detecting significant serial dependence in general (Weiß, 2011a), it cannot be recommended as an estimator for the actual value of the DAR(1) dependence parameter ϕ_1 . The estimator $\hat{\phi}_{1,C}$ based on Cohen’s κ , in contrast, performs reasonably well in any of the considered situations.

Example 5.3.2 (Bovine Data). For the Bovine sequence discussed in Section 3.4, we obtain the estimates 0.0806, 0.0250, 0.0009, ... for $\hat{\kappa}(k)$ according to (21), and 0.0807, 0.0260, 0.0024, ... for $\hat{\kappa}^*(k)$ according to (22). Using formula (7) in Weiß (2011a) (nominal size 0.05), we classify the κ -values for lags $k = 1, 2$ as being significantly different from 0 (critical values ± 0.0125 and ± 0.0123 , respectively).

Table 9Empirical correlation of simulated estimator $(\hat{\phi}_{1,C}, \hat{\phi}_{1,M})$.

Marginal: $\mathbf{p}_1 = (0.2, 0.2, 0.25, 0.35)^\top$					$\mathbf{p}_2 = (0.05, 0.1, 0.15, 0.7)^\top$			
$T \setminus \phi_1$	0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8
100	0.97	0.95	0.90	0.67	0.83	0.79	0.72	0.65
200	0.98	0.97	0.95	0.84	0.83	0.80	0.74	0.60
500	0.98	0.98	0.98	0.95	0.85	0.85	0.80	0.64
1000	0.98	0.99	0.98	0.98	0.86	0.87	0.85	0.78

**Fig. 2.** Power graphs for $\hat{\kappa}(1)$ with regard to the DAR(1) model, marginal distributions $\mathbf{p}_1$ (solid) and $\mathbf{p}_2$ (dashed).

From the above values for $\hat{\kappa}(k)$ and $\hat{\kappa}^*(k)$, we obtain the point estimates $\hat{\phi}_{1,C} \approx 0.0806$ and $\hat{\phi}_{1,M} \approx 0.0807$. [Corollary 5.3.1](#) now allows us to approximate their standard errors as 0.0068 in both cases. Using these standard errors and the asymptotic normality according to [Corollary 5.3.1](#), we may construct confidence intervals for ϕ_1 .

5.4. Testing for significant dependence

One of the main applications of the empirical autocorrelation function in real-valued time series analysis is to uncover a significantly nonzero autocorrelation. An analogous approach is also possible for categorical time series if using one of our κ -measures (or one of the measures for unsigned serial dependence as discussed later in Section 6), also see [Example 5.3.2](#) before. [Weiß \(2011a\)](#) considered tests of the null hypothesis of an i.i.d. categorical process, which were based on the statistics $\hat{\kappa}(k)$ or $\hat{\kappa}^*(k)$. The critical values of the considered κ -measures were computed by using their asymptotic properties (see (7) in [Weiß, 2011a](#)). In a simulation study, he investigated the empirical size of these tests and compared it to the nominal one. It turned out that $\hat{\kappa}$ showed a good performance in any of the considered situations, while this was true for $\hat{\kappa}^*$ only if $T \geq 500$. On the other hand, $\hat{\kappa}^*$ has the practical advantage that its asymptotic distribution (in the i.i.d. case) does not depend on the actual marginal distribution, see Theorem 2.4 in [Weiß \(2011a\)](#).

Concerning the power of these tests, we find some simulated values in Tables 6 and 8 in [Weiß \(2011a\)](#). But our new results in [Theorems 5.1.1](#) and [5.2.1](#) allow us to analyze the asymptotic power function directly, without the need to do simulations. This is illustrated in the following by considering the alternative of a DAR(1) model with dependence parameter ϕ_1 . In [Corollary 5.3.1](#), we already derived closed-form expressions for the asymptotic distribution of $\hat{\kappa}(1)$ and $\hat{\kappa}^*(1)$ in this case. From the simulation study in Section 5.3, we concluded that the asymptotic approximation is particularly good for $\hat{\kappa}(1)$, so we shall focus on this test statistic in the sequel. Selected asymptotic power graphs (nominal size 0.05) are shown in [Fig. 2](#).

It becomes clear that for long time series such as $T = 1000$, already small deviations from the i.i.d. model, like $\phi_1 \approx 0.05$, are successfully detected (remember that we even have $T = 8419$ in the example of Section 3.4). [Fig. 2](#) also shows that the given marginal distribution is of importance. The sensitivity of $\hat{\kappa}(1)$ to detect dependence is clearly better in the situation $\mathbf{p}_1$, where we have large dispersion. This behavior is plausible since in the case of low dispersion, i.e., if we have one dominant value in the range that attracts most of the probability mass, then already an i.i.d. process will show long runs of this symbol.

6. Empirical measures of unsigned serial dependence

For estimating the measures of unsigned serial dependence from (2) (with a fixed $k \geq 1$), we require all the estimators $\hat{p}_i(T)$ and $\hat{p}_{ij}(k, T)$, i.e., we define the process $(\mathbf{X}_t)_{\mathbb{Z}}$ by

$$\begin{aligned} \mathbf{X}_t &:= \mathbf{X}_t(k) := (X_{t,0}, \dots, X_{t,m}, X_{t,00}(k), X_{t,10}(k), \dots, X_{t,mm}(k))^\top \in \mathbb{R}^{(m+2)(m+1)} \\ \text{with } X_{t,i} &:= Y_{t,i} - p_i, \quad X_{t,ij}(k) := Y_{t,i}Y_{t-k,j} - p_{ij}(k). \end{aligned} \quad (23)$$

Obviously, each $\mathbf{X}_t$ has mean $\mathbf{0}$, and together with the above discussion, it becomes clear that $(\mathbf{X}_t)_{\mathbb{Z}}$ satisfies the conditions of the asymptotic result (9). Note that $\frac{1}{T} \sum_{t=1}^T X_{t,i} = \hat{p}_i(T) - p_i$ and $\frac{1}{T-k} \sum_{t=k+1}^T X_{t,ij}(k) = \hat{p}_{ij}(k, T) - p_{ij}(k)$, respectively.

Applying the asymptotic result (9) to the process $(\mathbf{X}_t)_\mathbb{Z}$ defined by formula (23), we obtain that $\sqrt{T} \cdot \left(\frac{1}{T} \sum_{t=1}^T \mathbf{X}_t \right)$ is asymptotically distributed according to $N(\mathbf{0}, \Sigma)$, where the covariance matrix Σ is a block matrix

$$\begin{aligned} \Sigma &= \left(\begin{array}{c|c} \sigma_{ij} & \sigma_{i;ab} \\ \hline \sigma_{ab;j} & \sigma_{ij;ab} \end{array} \right) \quad \text{with } \sigma_{ij} = p_j \cdot (\delta_{ij} - p_i) \cdot c, \\ \sigma_{i;ab} &= (\delta_{ia} - p_i) \cdot p_{ab}(k) + \sum_{n=1}^{k-1} (p_{aib}(n, k-n) - p_i p_{ab}(k)) \\ &\quad + (\delta_{ib} - p_i) \cdot p_{ab}(k) + \sum_{n=k+1}^{\infty} (p_{abi}(k, n-k) - p_i p_{ab}(k)) + \sum_{n=1}^{\infty} (p_{iab}(n, k) - p_i p_{ab}(k)), \\ \sigma_{ij;ab} &= (\delta_{ia} \cdot \delta_{jb} - p_{ab}(k)) \cdot p_{ij}(k) + \sum_{n=1}^{k-1} (p_{aibj}(n, k-n, n) + p_{iajb}(n, k-n, n) - 2p_{ij}(k)p_{ab}(k)) \\ &\quad + (\delta_{ib} \cdot p_{aj}(k, k) + \delta_{ja} \cdot p_{iab}(k, k) - 2p_{ij}(k)p_{ab}(k)) \\ &\quad + \sum_{n=k+1}^{\infty} (p_{abij}(k, n-k, k) + p_{ijab}(k, n-k, k) - 2p_{ij}(k)p_{ab}(k)). \end{aligned} \quad (24)$$

See Remark 5.1 for the used abbreviations. If the underlying process $(X_t)_\mathbb{Z}$ is even i.i.d., then the entries of Σ simplify to

$$\begin{aligned} \sigma_{i;ab} &= (\delta_{ia} + \delta_{ib} - 2p_i) \cdot p_a p_b, \\ \sigma_{ij;ab} &= \delta_{ia} \cdot \delta_{jb} \cdot p_i p_j + \delta_{ib} \cdot p_i p_j p_a + \delta_{ja} \cdot p_i p_a p_b - 3p_i p_j p_a p_b. \end{aligned} \quad (25)$$

The proof of formulas (24) and (25) is given in Appendix D.1.

6.1. Cramer's v

In view of the recommendation of Weiß and Göb (2008a), see formula (10) above, we consider the following estimator of Cramer's v according to (2):

$$\hat{v}^2(k) = \frac{1}{m} \cdot \sum_{i,j=0}^m \frac{(\hat{p}_{ij}(k, T) - \hat{p}_i(T)\hat{p}_j(T))^2}{\hat{p}_i(T)\hat{p}_j(T)}. \quad (26)$$

Note that again $\min_j \{\hat{p}_j(T)\} > 0$ is required to be able to compute this measure at all.

For analyzing the asymptotics of the measure (26), we shall proceed in two steps. First, we define the function $\mathbf{g}_v : \mathbb{R}^{(m+1)(m+2)} \rightarrow \mathbb{R}^{(m+1)^2}$ via

$$\mathbf{g}_{v,ij}(x_0, \dots, x_m, y_{00}, y_{10}, \dots, y_{mm}) = \frac{y_{ij} - x_i x_j}{\sqrt{x_i x_j}}. \quad (27)$$

We show that the transformed statistics $\mathbf{g}_v \left(\dots, \frac{1}{T} \sum_{t=1}^T Y_{t,i}, \dots, \frac{1}{T} \sum_{t=1}^T Y_{t,i} Y_{t-k,j}, \dots \right)$ follow asymptotically a certain type of multivariate normal distribution. Then in the second step, we analyze the asymptotic distribution of the resulting sums of squares of the components of these statistics, which are very closely related to the measure (26). Since the involved expressions turn out to be quite complex, we exemplify this procedure only for an underlying i.i.d. process, although it would be done analogously for the general NDARMA case. This i.i.d. situation is also the main application we have in mind, because it will allow us to identify significant serial dependence in a given categorical time series, in analogy to the approach for the measures of signed serial dependence in Weiß (2011a) and in Section 5 above. But it seems to be the first time for the case of measures of unsigned serial dependence, also see Section 3.2 in Bagnato et al. (2012).

Theorem 6.1.1 (Cramer's v). Let $X_1, \dots, X_T$ be a segment from an i.i.d. categorical process $(X_t)_\mathbb{Z}$. Let $k \in \mathbb{N}$. Then the distribution of the empirical Cramer's v of formula (26), $\hat{v}(k)$, is asymptotically approximated by the following relation:

$$T \cdot m \cdot \hat{v}^2(k) \underset{\text{appr.}}{\sim} \chi_{m^2}^2,$$

a χ^2 -distribution with m^2 degrees of freedom.

The proof of Theorem 6.1.1 is provided by Appendix D.2. Since the mean of a $\chi_{m^2}^2$ -distributed random variable equals $m^2 > 0$, we recognize that at least for small T , $\hat{v}(k)$ is problematic as an estimator of $v(k)$ in terms of bias. But this observation does not affect the proposed use of $\hat{v}(k)$ for uncovering significant serial dependence. Theorem 6.1.1 implies to refer to the

Table 10

Empirical rejection rates for Cramer's $\hat{v}(1)$, see (28), with regard to the 95%-quantile of the χ^2_9 -distribution, i.e., with critical value $2.3748/\sqrt{T}$.

Marginal:		$\mathbf{p}_1$	$\mathbf{p}_2$	Marginal:		$\mathbf{p}_1$	$\mathbf{p}_2$
Model	T	r.r.	r.r.	Model	T	r.r.	r.r.
i.i.d.	100	0.050	0.056	DMA(1), $\varphi_1 = 0.25$	100	0.489	0.517
	200	0.048	0.053		200	0.879	0.847
	500	0.052	0.050		500	1.000	0.999
	1000	0.048	0.051		1000	1.000	1.000

measured serial dependence at lag k as being significantly different from 0 if

$$\hat{v}(k) > \frac{1}{\sqrt{T \cdot m}} \cdot \sqrt{q_{1-\alpha}}, \quad (28)$$

where $q_{1-\alpha}$ denotes the $(1 - \alpha)$ -quantile of the $\chi^2_{m^2}$ -distribution. Note that a similar rule was used by Bagnato et al. (2012) for their autodependence function (but without a theoretical justification).

6.2. Using Cramer's v to uncover significant dependence

The goodness of rule (28) for time series of finite length T as well as the goodness of the whole approximation of Theorem 6.1.1 was investigated in a simulation study in analogy to Section 3; detailed results are available upon request. The quantile plots for the i.i.d. simulations (not shown) indicate that the distribution of $T \cdot m \cdot \hat{v}^2(1)$ is always very close to the $\chi^2_{m^2}$ -distribution for marginal $\mathbf{p}_1$, while we observe slight deviations for marginal $\mathbf{p}_2$ if $T \leq 200$. These observations also become manifest in the rejection rates shown in Table 10 for the i.i.d. case, where we have slightly increased rejection rates (with regard to the nominal 5% level) for marginal $\mathbf{p}_2$ if $T \leq 200$.

Let us now consider the power of the decision rule (28). For our DAR(1) models with $\phi_1 = 0.4, 0.8$, this rule leads nearly always to a rejection of the i.i.d. hypothesis, so we do not show the corresponding results. But the rejection rates for the DMA(1) model are summarized in Table 10, where we have the low dependence level $v(1) = \varphi_1(1 - \varphi_1) = 0.1875$ according to Example 2.3 in Weiß (2011b). It becomes clear that, independent of the marginal distribution, time series of length $T \geq 200$ are necessary to reliably detect such a violation of the i.i.d. assumption. Comparing this with the corresponding results in Tables 6–8 in Weiß (2011a), we recognize that the κ -measures are more powerful with regard to such a DMA(1)-dependence.

Example 6.2.1 (Bovine Data). For the Bovine sequence discussed in Section 3.4, we have 0.1134, 0.0445, 0.0281, 0.0222, ... for $\hat{v}(k)$, see Example 6.1 in Weiß and Göb (2008a). Our new results from Section 6.1, especially formula (28), now lead to the critical value 0.0259 (nominal size 0.05), i.e., $\hat{v}(1)$, $\hat{v}(2)$ and $\hat{v}(3)$ can now be identified as being significantly larger than 0.

6.3. Goodman and Kruskal's τ

Next, we repeat the procedure of Appendix D.2 for the empirical version of Goodman and Kruskal's τ according to (2), defined as

$$\hat{\tau}^2(k) = \sum_{i,j=0}^m \frac{(\hat{p}_{ij}(k, T) - \hat{p}_i(T)\hat{p}_j(T))^2}{\hat{p}_j(T)(1 - s_2(\hat{\mathbf{p}}(T)))}. \quad (29)$$

Note that again $\min_j \{\hat{p}_j(T)\} > 0$ is required to be able to compute this measure at all. Instead of $\mathbf{g}_v$, we now consider the function $\mathbf{g}_\tau : \mathbb{R}^{(m+1)(m+2)} \rightarrow \mathbb{R}^{(m+1)^2}$, defined via

$$\mathbf{g}_{\tau,ij}(x_0, \dots, x_m, y_{00}, y_{10}, \dots, y_{mm}) = \frac{y_{ij} - x_i x_j}{\sqrt{x_j \left(1 - \sum_{r=0}^m x_r^2\right)}}. \quad (30)$$

As we shall see in Appendix D.3, it turns out that the distribution of $\hat{\tau}(k)$ cannot be approximated based on a simple χ^2 -distribution, but knowledge about the true marginal distribution $\mathbf{p}$ is necessary, which makes $\hat{\tau}(k)$ less attractive than $\hat{v}(k)$ from a practical point of view.

Theorem 6.3.1 (Goodman and Kruskal's τ). Let $X_1, \dots, X_T$ be a segment from an i.i.d. categorical process $(X_t)_{\mathbb{Z}}$. Let $k \in \mathbb{N}$. Let

$$\Sigma_\tau = \left(\frac{p_r}{1 - s_2(\mathbf{p})} \cdot (\delta_{rt} - p_t) (\delta_{su} - \sqrt{p_s p_u}) \right)_{rs,tu} \in \mathbb{R}^{(m+1)^2 \times (m+1)^2},$$

Table 11

Empirical rejection rates for Goodman and Kruskal's $\hat{\tau}(1)$, see [Theorem 6.3.1](#), with regard to the 95%-quantile of the approximate distribution, i.e., with critical value $2.3856/\sqrt{T}$ for $\mathbf{p}_1$ and $2.4916/\sqrt{T}$ for $\mathbf{p}_2$.

Marginal:		$\mathbf{p}_1$	$\mathbf{p}_2$	Marginal:		$\mathbf{p}_1$	$\mathbf{p}_2$
Model	T	r.r.	r.r.	Model	T	r.r.	r.r.
i.i.d.	100	0.054	0.050	DMA(1), $\varphi_1 = 0.25$	100	0.491	0.436
	200	0.050	0.050		200	0.877	0.782
	500	0.053	0.049		500	1.000	0.997
	1000	0.050	0.050		1000	1.000	1.000

and let $\lambda_1, \dots, \lambda_l$ denote the nonzero distinct eigenvalues of Σ_τ with algebraic multiplicities $r_1, \dots, r_l$, respectively. Then the distribution of the empirical Goodman and Kruskal's τ of formula (29), $\hat{\tau}(k)$, is asymptotically approximated by the following relation: $T \cdot \hat{\tau}^2(k)$ shows approximately the same distribution as $\sum_{i=1}^l \lambda_i \cdot Q_i$, where $Q_1, \dots, Q_l$ are independent random variables with Q_i being χ^2 -distributed with r_i degrees of freedom.

Besides the above-mentioned disadvantage of requiring knowledge about $\mathbf{p}$, also the computation of quantiles for the distribution in [Theorem 6.3.1](#) is more difficult than in the χ^2 -case (below, we use the R-package `CompQuadForm` for this purpose). In view of these results, it appears that for practitioners, Cramer's v is preferable to τ as a measure of unsigned serial dependence.

Nevertheless, we also analyze the finite-sample performance of the approximation according to [Theorem 6.3.1](#). In a first step, we numerically compute the nonzero distinct eigenvalues of Σ_τ ; we obtain 0.2721, 0.3103, 0.4176 for $\mathbf{p}_1$ and 0.1254, 0.2503, 0.6243 for $\mathbf{p}_2$, always with algebraic multiplicity 3. Then, we use the function `davies` or `imhof` from the R-package `CompQuadForm` (together with `uniroot`) to determine the approximate 95%-quantiles of $\sqrt{T} \cdot \hat{\tau}(1)$ as 2.3856 and 2.4916, respectively. The resulting empirical rejection rates are summarized in [Table 11](#). Comparing these values with the ones of [Table 10](#), we recognize that in the i.i.d. situation, $\hat{\tau}(1)$ is less influenced by varying dispersion than $\hat{v}(1)$, but which is reasonable if we look at the definition of these measures: $\hat{\tau}(1)$ is “adjusted” by $1 - s_2(\hat{\mathbf{p}}(T))$, which is essentially equal to the empirical Gini index. On the other hand, it shows clearly less power than $\hat{v}(1)$ in the situation $\mathbf{p}_2$ with moderate dispersion.

7. Conclusions and future research

We showed that an NDARMA process is φ -mixing. This is a useful property for deriving the asymptotic distribution of statistics being related to such a process, which was exemplified by several examples concerning marginal or bivariate characteristics of NDARMA processes. For estimating the marginal dispersion, we analyzed empirical versions of Gini index and entropy. Especially for the Gini index, we obtained asymptotic expressions that approximate the finite-sample characteristics quite well. As a representative for goodness-of-fit tests, we considered the well-known Pearson statistic and derived a simple expression for its asymptotic distribution. Then, we considered empirical measures of signed and unsigned serial dependence. We analyzed the κ -measures of signed dependence for the case of general NDARMA models and for the particular DAR(1) model. Especially the normal approximation for Cohen's κ works rather well also for short time series. It might be used for quantifying the uncertainty of estimates, or for power analysis when testing the null hypothesis of i.i.d. data. Among the two considered measures of unsigned dependence, if applied to i.i.d. data, Cramer's v turned to be the best choice, both concerning the simplicity of the asymptotic approximation as well as its power to detect dependence.

A future research project should provide a more detailed empirical analysis of the considered measures of (signed) serial dependence, where, e. g., the power to detect serial dependence is also analyzed with regard to non-NDARMA models. Also alternative types of goodness-of-fit statistics may be considered, and especially the effect of estimated parameters (e. g., concerning the parameter c necessary for the Pearson statistic) should be investigated. Finally, the above mixing properties rely on the finiteness of the categorical range. It might be interesting to analyze if certain mixing properties could also be derived if the range is countably infinite.

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Appendix A. Proofs of Section 2.2

The proofs of Section 2.2 are taken from Section 11.2 of [Weiß \(2009\)](#).

A.1. Proof of Theorem 2.2.1

Lemma. Let $(\mathbf{Z}_t)_{\mathbb{Z}}$ be the Markov representation of an NDARMA process according to formula (7), and let $p + q \geq 1$. Denote the transition probabilities $p_{\mathbf{a}|\mathbf{b}}(k) := P(\mathbf{Z}_t = \mathbf{a} \mid \mathbf{Z}_{t-k} = \mathbf{b})$. Then $p_{\mathbf{a}|\mathbf{b}}(p + q + 1) > 0$ for all $\mathbf{a}, \mathbf{b} \in \mathcal{V}^{\max\{p,1\}+q}$.

Proof. Let $\mathbf{a}, \mathbf{b} \in \mathcal{V}^{\max\{p,1\}+q}$ be arbitrary vectors, and denote

$$\mathbf{a} = (a_0; \dots, a_{p-1}, b_0, \dots, b_{q-1})^\top, \quad \mathbf{b} = (a_{p+q+1}; \dots, a_{2p+q}, b_{p+q+1}, \dots, b_{p+2q})^\top.$$

Then

$$\begin{aligned} p_{\mathbf{a}|\mathbf{b}}(p + q + 1) &= P(\mathbf{Z}_t = \mathbf{a} \mid \mathbf{Z}_{t-(p+q+1)} = \mathbf{b}) \\ &= \sum_{\substack{a_{\max\{p,1\}, \dots, a_{p+q}}, \\ b_q, \dots, b_{p+q}}} P(X_t = a_0, \dots, X_{t-(p+q)} = a_{p+q}, \epsilon_t = b_0, \dots, \epsilon_{t-(p+q)} = b_{p+q} \mid \mathbf{Z}_{t-(p+q+1)} = \mathbf{b}) \\ &= \sum_{\substack{a_{\max\{p,1\}, \dots, a_{p+q}}, \\ b_q, \dots, b_{p+q}}} \prod_{j=0}^{p+q} \left(P(X_{t-j} = a_j \mid X_{t-j-1} = a_{j+1}, \dots, \epsilon_{t-j} = b_j, \dots) \right. \\ &\quad \left. \cdot P(\epsilon_{t-j} = b_j \mid X_{t-j-1} = a_{j+1}, \dots, \epsilon_{t-j-1} = b_{j+1}, \dots) \right). \end{aligned}$$

Because of the independence assumptions concerning $(\epsilon_t)_{\mathbb{Z}}$, see Definition 2.1.1,

$$P(\epsilon_{t-j} = b_j \mid X_{t-j-1} = a_{j+1}, \dots, \epsilon_{t-j-1} = b_{j+1}, \dots) = p_{b_j}.$$

So the above summands of $p_{\mathbf{a}|\mathbf{b}}(p + q + 1)$ simplify to

$$\prod_{j=0}^{p+q} \left(p_{b_j} \cdot \left(\sum_{r=1}^p \phi_r \cdot \delta_{a_j, a_{j+r}} + \sum_{s=0}^q \varphi_s \cdot \delta_{a_j, b_{j+s}} \right) \right).$$

They are obviously nonnegative, as a product of nonnegative factors. So if there exists at least one summand with only positive factors, then $p_{\mathbf{a}|\mathbf{b}}(p + q + 1) > 0$.

Case 1: $p = 0, q \geq 1$. Consider the summand with $a_1 := b_{q+1}, \dots, a_q := b_{2q}, b_q := a_0$, then

$$\sum_{s=0}^q \varphi_s \cdot \delta_{a_j, b_{j+s}} \geq \varphi_q \cdot \delta_{a_j, b_{j+q}} = \varphi_q > 0 \quad \text{for all } j = 0, \dots, q.$$

Case 2: $p \geq 1, q = 0$. Consider the summands with $b_1 := a_1, \dots, b_p := a_p := a_0$, then

$$\sum_{r=1}^p \phi_r \cdot \delta_{a_j, a_{j+r}} + \varphi_0 \cdot \delta_{a_j, b_j} \geq \begin{cases} \phi_p \cdot \delta_{a_0, a_p} = \phi_p > 0 & \text{if } j = 0, \\ \varphi_0 \cdot \delta_{a_j, b_j} = \varphi_0 > 0 & \text{if } j = 1, \dots, p. \end{cases}$$

Case 3: $p, q \geq 1$. Consider the summands with $b_q := a_0, \dots, b_{p+q} := a_p, a_{p+1} := b_{p+q+1}, \dots, a_{p+q} := b_{p+2q}$, and with arbitrary $a_p \in \mathcal{V}$. Then

$$\sum_{r=1}^p \phi_r \cdot \delta_{a_j, a_{j+r}} + \sum_{s=0}^q \varphi_s \cdot \delta_{a_j, b_{j+s}} \geq \varphi_q \cdot \delta_{a_j, b_{j+q}} = \varphi_q > 0$$

for all $j = 0, \dots, p + q$. $\square$

The lemma implies that $(\mathbf{Z}_t)_{\mathbb{Z}}$ is primitive. So according to Theorem 1.4 in Seneta (1983), it is also an ergodic Markov chain. As a consequence, $(\mathbf{Z}_t)_{\mathbb{Z}}$ is also ψ - and φ -mixing (Blum et al., 1963), with exponentially decreasing weights $f_n = a \cdot \rho^n$ with $a > 0$ and $0 < \rho < 1$ (Billingsley, 1968, p. 167f). In particular, $\sum_{n \in \mathbb{N}} f_n^{1/2} < \infty$.

Denote $[X_t]_a^b := (X_b, \dots, X_a)$ for all $a, b \in \mathbb{Z}$ with $b > a$. Let $k, l \geq \tilde{p} := \max\{1, p\}$, $\mathbf{a}_1 \in \mathcal{V}^k, \mathbf{a}_2 \in \mathcal{V}^l$ and $n \in \mathbb{N}$ be given arbitrarily. Then

$$\begin{aligned} &|P([X_t]_n^{n+k-1} = \mathbf{a}_1, [X_t]_{n-l+1}^0 = \mathbf{a}_2) - P([X_t]_n^{n+k-1} = \mathbf{a}_1) \cdot P([X_t]_{n-l+1}^0 = \mathbf{a}_2)| \\ &= \left| \sum_{\mathbf{b}_1, \mathbf{b}_2} \left(P([X_t]_n^{n+k-1} = \mathbf{a}_1, [\epsilon_t]_{n-q+\tilde{p}}^{n+k-1} = \mathbf{b}_1, [X_t]_{n-l+1}^0 = \mathbf{a}_2, [\epsilon_t]_{n-l-q+\tilde{p}+1}^0 = \mathbf{b}_2) \right. \right. \\ &\quad \left. \left. - P([X_t]_n^{n+k-1} = \mathbf{a}_1, [\epsilon_t]_{n-q+\tilde{p}}^{n+k-1} = \mathbf{b}_1) \cdot P([X_t]_{n-l+1}^0 = \mathbf{a}_2, [\epsilon_t]_{n-l-q+\tilde{p}+1}^0 = \mathbf{b}_2) \right) \right| \end{aligned}$$

$$= \left| \sum_{\mathbf{b}_1, \mathbf{b}_2} \left(P(\mathbf{Z}_{n+k-1} = \mathbf{z}_{n+k-1}, \dots, \mathbf{Z}_{n+\tilde{p}-1} = \mathbf{z}_{n+\tilde{p}-1}, \mathbf{Z}_0 = \mathbf{z}_0, \dots, \mathbf{Z}_{-l+\tilde{p}} = \mathbf{z}_{-l+\tilde{p}}) \right. \right. \\ \left. \left. - P(\mathbf{Z}_{n+k-1} = \mathbf{z}_{n+k-1}, \dots, \mathbf{Z}_{n+\tilde{p}-1} = \mathbf{z}_{n+\tilde{p}-1}) \cdot P(\mathbf{Z}_0 = \mathbf{z}_0, \dots, \mathbf{Z}_{-l+\tilde{p}} = \mathbf{z}_{-l+\tilde{p}}) \right) \right|,$$

where the summation is done for all appropriate $\mathbf{b}_1 \in \mathcal{V}^{q+k-\tilde{p}}$, $\mathbf{b}_2 \in \mathcal{V}^{q+l-\tilde{p}}$. Furthermore, the vectors $\mathbf{z}_j$ in the last step are uniquely determined by the definition of the Markov chain representation $(\mathbf{Z}_t)_{\mathbb{Z}}$ according to formula (7). $(\mathbf{Z}_t)_{\mathbb{Z}}$ is ψ - and φ -mixing with the exponentially decreasing weights denoted by $(g_n)_{\mathbb{N}}$. So by definition, this implies the inequality

$$\begin{aligned} & |P([X_t]_n^{n+k-1} = \mathbf{a}_1, [X_t]_{-l+1}^0 = \mathbf{a}_2) - P([X_t]_n^{n+k-1} = \mathbf{a}_1) \cdot P([X_t]_{-l+1}^0 = \mathbf{a}_2)| \\ & \leq \sum_{\mathbf{b}_1, \mathbf{b}_2} g_{n+\tilde{p}-1} \cdot P(\mathbf{Z}_{n+k-1} = \mathbf{z}_{n+k-1}, \dots, \mathbf{Z}_{n+\tilde{p}-1} = \mathbf{z}_{n+\tilde{p}-1}) \cdot P(\mathbf{Z}_0 = \mathbf{z}_0, \dots, \mathbf{Z}_{-l+\tilde{p}} = \mathbf{z}_{-l+\tilde{p}}) \\ & = g_{n+\tilde{p}-1} \cdot P([X_t]_n^{n+k-1} = \mathbf{a}_1) \cdot P([X_t]_{-l+1}^0 = \mathbf{a}_2). \end{aligned}$$

The proof is easily adapted if $k < p$ or $l < p$: we simply have to extend the above summation by adding the missing $p - k$ or $p - l$ X_t 's, respectively. So $(X_t)_{\mathbb{Z}}$ is ψ -mixing with the weights $f_n := g_{n+\tilde{p}-1}$.

A.2. Proof of formula (11)

Let $\phi_{\bullet} := \sum_{j=1}^p \phi_j$. According to (5), we obtain for $k > \max\{p, q\}$ that

$$\kappa(k) = \sum_{j=1}^p \phi_j \cdot \kappa(k-j) \leq \phi_{\bullet} \cdot \max_{j=1, \dots, p} \kappa(k-j).$$

Let $\nu := \sqrt[p]{\phi_{\bullet}}$ and $r := \max\{p, q\} + 1$, where $\nu < 1$ according to Definition 2.1.1. For any $k = r + n \cdot p + s$ with $0 \leq s \leq p-1$ and $n \in \mathbb{N}_0$, one shows that there exists a constant $d > 0$ such that

$$\kappa(k) \leq \phi_{\bullet}^{n+1} = \nu^{n \cdot p + p} \leq \nu^{n \cdot p + s} = d^r \cdot \nu^k,$$

where $d := \frac{1}{\nu}$ for $p > 0$. From this inequality, it follows for $p \geq 1$ that

$$\text{there exists a constant } a > 0 \text{ such that } \kappa(k) \leq a \cdot \nu^k \text{ for all } k \in \mathbb{N}. \quad (\text{A.1})$$

Since $\nu < 1$ according to Definition 2.1.1, this always guarantees that $\sum_{k=1}^{\infty} \kappa(k) < \infty$.

Appendix B. Proofs of Section 3

B.1. Proof of formula (14)

According to formula (10), we have $T^{-1/2} \cdot \sum_{t=1}^T \mathbf{X}_t = \sqrt{T} \cdot (\hat{\mathbf{p}}(T) - \mathbf{p})$. Furthermore, see (6),

$$E[X_{k,i} X_{0,j}] = \text{Cov}[Y_{k,i}, Y_{0,j}] = p_{ij}(k) - p_i p_j = p_j \cdot (\delta_{ij} - p_i) \cdot \kappa(k),$$

which is also equal to $E[X_{0,i} X_{k,j}]$. Hence,

$$\sigma_{ij} = E[X_{0,i} X_{0,j}] + \sum_{k=1}^{\infty} (E[X_{0,i} X_{k,j}] + E[X_{k,i} X_{0,j}]) = p_j \cdot (\delta_{ij} - p_i) \cdot c.$$

Applying Theorem 2.2.1 and the asymptotic result (9), assertion (14) follows.

B.2. Proof of Theorem 3.1.1

Consider the function $g_G : \mathbb{R}^{m+1} \rightarrow \mathbb{R}$ defined by

$$g_G(x_0, \dots, x_m) := 1 - \sum_{j=0}^m x_j^2.$$

Then $\frac{\partial}{\partial x_j} g_G = -2x_j$. So the respective gradient, evaluated in $\mathbf{p}$, is given by

$$\mathbf{D}_G = -2\mathbf{p}^T.$$

Applying the asymptotic result (14) and the delta theorem (Section 3.3 in Serfling, 1980), we obtain that

$$\sqrt{T} \cdot (g_G(\hat{\mathbf{p}}(T)) - g_G(\mathbf{p})) \xrightarrow{D} N(\mathbf{0}, c \cdot \mathbf{D}_G \Sigma \mathbf{D}_G^\top),$$

where

$$\mathbf{D}_G \Sigma \mathbf{D}_G^\top = 4\mathbf{p}^\top \Sigma \mathbf{p} = 4 \sum_{i,j=0}^m p_i p_j \cdot \sigma_{ij} = 4 (s_3(\mathbf{p}) - s_2^2(\mathbf{p})).$$

B.3. Proof of Theorem 3.2.1

Consider the function $g_E : \mathbb{R}^{m+1} \rightarrow \mathbb{R}$ defined by

$$g_E(x_0, \dots, x_m) := - \sum_{j=0}^m x_j \ln x_j.$$

Then $\frac{\partial}{\partial x_j} g_E = -\ln x_j - 1$. So the respective gradient, evaluated in $\mathbf{p}$, is given by

$$\mathbf{D}_E = -(1 + \ln p_0, \dots, 1 + \ln p_m).$$

Applying the asymptotic result (14) and the delta theorem (Section 3.3 in Serfling, 1980), we obtain that

$$\sqrt{T} \cdot (g_E(\hat{\mathbf{p}}(T)) - g_E(\mathbf{p})) \xrightarrow{D} N(\mathbf{0}, c \cdot \mathbf{D}_E \Sigma \mathbf{D}_E^\top),$$

where

$$\begin{aligned} \mathbf{D}_E \Sigma \mathbf{D}_E^\top &= \sum_{i,j=0}^m (1 + \ln p_i)(1 + \ln p_j) \cdot \sigma_{ij} \\ &= \sum_{i,j=0}^m (1 + \ln p_i)(1 + \ln p_j) \cdot p_i \cdot (\delta_{ij} - p_j) \\ &= \sum_{i=0}^m p_i \cdot (1 + \ln p_i)^2 - \left(\sum_{i=0}^m p_i \cdot (1 + \ln p_i) \right)^2 \\ &= 1 + 2 \sum_{i=0}^m p_i \ln p_i + \sum_{i=0}^m p_i \cdot (\ln p_i)^2 - \left(1 + \sum_{i=0}^m p_i \ln p_i \right)^2. \end{aligned}$$

B.4. Proof of formula (17)

We have

$$\begin{aligned} E[\hat{p}_i(T)^2] &= \frac{1}{T^2} \cdot \sum_{s,t=1}^T E[Y_{t,i} Y_{s,i}] = \frac{1}{T^2} \cdot \sum_{t=1}^T p_i + \frac{2}{T^2} \cdot \sum_{s=1}^{T-1} \sum_{t=s+1}^T p_{ii}(t-s) \\ &= \frac{p_i}{T} + \frac{2}{T^2} \cdot \sum_{k=1}^{T-1} \sum_{s=1}^{T-k} p_{ii}(k) \\ &= \frac{p_i}{T} + \frac{2}{T^2} \cdot \sum_{k=1}^{T-1} (T-k) \cdot (p_i^2 + p_i(1-p_i) \cdot \kappa(k)) \\ &= \frac{p_i}{T} + \frac{2p_i^2}{T^2} \cdot \frac{(T-1)T}{2} + \frac{2p_i(1-p_i)}{T} \cdot \sum_{k=1}^{T-1} \left(1 - \frac{k}{T}\right) \cdot \kappa(k) \\ &= p_i^2 + \frac{p_i(1-p_i)}{T} \cdot \left(1 + 2 \sum_{k=1}^{T-1} \left(1 - \frac{k}{T}\right) \cdot \kappa(k)\right) \leq p_i^2 + \frac{p_i(1-p_i)}{T} \cdot c. \end{aligned}$$

Hence,

$$\begin{aligned} E[\hat{v}_G(X)] &= \frac{m+1}{m} \cdot \frac{T}{T-1} \cdot \left(1 - \sum_{i=0}^m E[\hat{p}_i(T)^2]\right) \\ &= v_G(X) \cdot \frac{T}{T-1} \cdot \left(1 - \frac{1}{T} \cdot \left(1 + 2 \sum_{k=1}^{T-1} \left(1 - \frac{k}{T}\right) \cdot \kappa(k)\right)\right) \geq v_G(X) \cdot \frac{T-c}{T-1}. \end{aligned}$$

Appendix C. Proofs of Section 5

C.1. Proof of Theorems 5.1.1 and 5.2.1

We shall apply the delta theorem (Serfling, 1980, Section 3.3) to the asymptotic result (19). For this purpose, we define the functions $g_C, g_M : \mathbb{R}^{2(m+1)} \rightarrow \mathbb{R}$ via

$$g_C(x_0, \dots, x_m, y_0, \dots, y_m) := \frac{1 - \sum_{r=0}^m y_r}{1 - \sum_{r=0}^m x_r^2}, \quad g_M(x_0, \dots, x_m, y_0, \dots, y_m) := \sum_{r=0}^m \frac{y_r}{x_r}.$$

Then

$$\frac{\partial}{\partial x_j} g_C = 2x_j \cdot \frac{1 - \sum_{r=0}^m y_r}{\left(1 - \sum_{r=0}^m x_r^2\right)^2}, \quad \frac{\partial}{\partial y_j} g_C = \frac{-1}{1 - \sum_{r=0}^m x_r^2}, \quad \frac{\partial}{\partial x_j} g_M = -\frac{y_j}{x_j^2}, \quad \frac{\partial}{\partial y_j} g_M = \frac{1}{x_j}.$$

So the respective gradients, evaluated in $(p_0, \dots, p_m, p_{00}(k), \dots, p_{mm}(k))^T$, are given by

$$\mathbf{D}_C = \frac{1}{1 - s_2(\mathbf{p})} \cdot \left(2p_0 \cdot \frac{1 - \sum_{j=0}^m p_{jj}(k)}{1 - s_2(\mathbf{p})}, \dots, 2p_m \cdot \frac{1 - \sum_{j=0}^m p_{jj}(k)}{1 - s_2(\mathbf{p})}, -1, \dots, -1 \right),$$

$$\mathbf{D}_M = \left(-\frac{p_{00}(k)}{p_0^2}, \dots, -\frac{p_{mm}(k)}{p_m^2}, \frac{1}{p_0}, \dots, \frac{1}{p_m} \right).$$

Computing

$$\mathbf{D}\Sigma\mathbf{D}^T = \sum_{i,j=0}^m d_i d_j \cdot \sigma_{ij} + 2 \cdot \sum_{i,b=0}^m d_i d_{bb} \cdot \sigma_{i;bb} + \sum_{a,b=0}^m d_{aa} d_{bb} \cdot \sigma_{aa;bb}$$

with Σ from (19), we obtain

$$\begin{aligned} \mathbf{D}_C \Sigma \mathbf{D}_C^T &= 4 \cdot \frac{\left(1 - \sum_{r=0}^m p_{rr}(k)\right)^2}{(1 - s_2(\mathbf{p}))^4} \cdot \sum_{i,j=0}^m p_i p_j \cdot \sigma_{ij} \\ &\quad - 4 \cdot \frac{1 - \sum_{r=0}^m p_{rr}(k)}{(1 - s_2(\mathbf{p}))^3} \cdot \sum_{i,b=0}^m p_i \cdot \sigma_{i;bb} + \frac{1}{(1 - s_2(\mathbf{p}))^2} \cdot \sum_{a,b=0}^m \sigma_{aa;bb} \\ &= 4c \cdot \frac{\left(1 - \sum_{r=0}^m p_{rr}(k)\right)^2}{(1 - s_2(\mathbf{p}))^4} \cdot (s_3(\mathbf{p}) - s_2^2(\mathbf{p})) \\ &\quad - 4 \cdot \frac{1 - \sum_{r=0}^m p_{rr}(k)}{(1 - s_2(\mathbf{p}))^3} \cdot \sum_{i,b=0}^m p_i \cdot \sigma_{i;bb} + \frac{1}{(1 - s_2(\mathbf{p}))^2} \cdot \sum_{a,b=0}^m \sigma_{aa;bb}, \end{aligned}$$

and

$$\begin{aligned} \mathbf{D}_M \Sigma \mathbf{D}_M^T &= \sum_{i,j=0}^m \frac{p_{ii}(k)p_{jj}(k)}{p_i^2 p_j^2} \cdot \sigma_{ij} - 2 \cdot \sum_{i,b=0}^m \frac{p_{ii}(k)}{p_i^2 p_b} \cdot \sigma_{i;bb} + \sum_{a,b=0}^m \frac{\sigma_{aa;bb}}{p_a p_b} \\ &= c \cdot \sum_{i=0}^m \frac{p_{ii}(k)^2}{p_i^3} - c \cdot \left(\sum_{i=0}^m p_{i|i}(k) \right)^2 - 2 \cdot \sum_{i,b=0}^m \frac{p_{ii}(k)}{p_i^2 p_b} \cdot \sigma_{i;bb} + \sum_{a,b=0}^m \frac{\sigma_{aa;bb}}{p_a p_b}. \end{aligned}$$

For the case of an i.i.d. process, we have $p_{ij}(k) = p_i p_j$ and $c = 1$. So together with the simplified expressions from formula (20), we obtain

$$\begin{aligned}
 \mathbf{D}_C \Sigma \mathbf{D}_C^\top &= 4 \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} - \frac{4}{(1 - s_2(\mathbf{p}))^2} \cdot \sum_{i,b=0}^m p_i \cdot \sigma_{i;bb} + \frac{1}{(1 - s_2(\mathbf{p}))^2} \cdot \sum_{a,b=0}^m \sigma_{aa;bb} \\
 &= 4 \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} - \frac{8}{(1 - s_2(\mathbf{p}))^2} \cdot \sum_{i,b=0}^m p_i \cdot (\delta_{ib} - p_i) p_b^2 \\
 &\quad + \frac{1}{(1 - s_2(\mathbf{p}))^2} \cdot \sum_{a,b=0}^m (\delta_{ab} \cdot p_a^2 \cdot (1 + 2p_a) - 3p_a^2 p_b^2) \\
 &= 4 \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} - 8 \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} + \frac{s_2(\mathbf{p}) + 2s_3(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} - \frac{3 \cdot s_2^2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} \\
 &= \frac{-2s_3(\mathbf{p}) + s_2^2(\mathbf{p}) + s_2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2} = 1 - \frac{1 + 2s_3(\mathbf{p}) - 3s_2(\mathbf{p})}{(1 - s_2(\mathbf{p}))^2}, \tag{C.1}
 \end{aligned}$$

and

$$\begin{aligned}
 \mathbf{D}_M \Sigma \mathbf{D}_M^\top &= c \cdot 1 - c \cdot 1 - 2 \cdot \sum_{i,b=0}^m \frac{\sigma_{i;bb}}{p_b} + \sum_{a,b=0}^m \frac{\sigma_{aa;bb}}{p_a p_b} \\
 &= 0 - 4 \cdot \sum_{i,b=0}^m (\delta_{ib} - p_i) p_b + \sum_{a,b=0}^m (\delta_{ab} \cdot (1 + 2p_a) - 3p_a p_b) \\
 &= 0 + \sum_{a=0}^m (1 + 2p_a) - 3 \cdot \sum_{a,b=0}^m p_a p_b = m. \tag{C.2}
 \end{aligned}$$

C.2. Proof of Corollary 5.3.1

To compute the covariance matrix Σ of formula (19), we need expressions for the joint probabilities $p_{ab}(k)$, $p_{abc}(k, l)$, $p_{abcd}(k, l, m)$. A DAR(1) process is a Markov chain with $\kappa(k) = \phi_1^k$ according to (5), also see Example 2.3 in Weiß (2011b). Using (6), we obtain that

$$\begin{aligned}
 p_{abcd}(k, l, m) &= p_{a|b}(k) \cdot p_{b|c}(l) \cdot p_{c|d}(m) \cdot p_d \\
 p_{abc}(k, l) &= p_{a|b}(k) \cdot p_{b|c}(l) \cdot p_c \\
 p_{ab}(k) &= p_{a|b}(k) \cdot p_b \quad \text{with } p_{a|b}(k) = p_a + (\delta_{ab} - p_a) \cdot \phi_1^k.
 \end{aligned}$$

For the covariance matrix Σ of formula (19) with fixed $k = 1$, we obtain

$$\begin{aligned}
 \sigma_{i;bb} &= 2(\delta_{ib} - p_i) \cdot p_{b|b}(1) \cdot p_b + \sum_{n=2}^{\infty} (p_{b|b}(1) \cdot p_{b|i}(n-1) \cdot p_i - p_i \cdot p_{b|b}(1) \cdot p_b) \\
 &\quad + \sum_{n=1}^{\infty} (p_{i|b}(n) \cdot p_{b|b}(1) \cdot p_b - p_i \cdot p_{b|b}(1) \cdot p_b) \\
 &= 2(\delta_{ib} - p_i) \cdot p_{b|b}(1) \cdot p_b + p_i \cdot p_{b|b}(1) \cdot (\delta_{ib} - p_b) \cdot \sum_{n=2}^{\infty} \phi_1^{n-1} + p_{b|b}(1) \cdot p_b \cdot (\delta_{ib} - p_i) \cdot \sum_{n=1}^{\infty} \phi_1^n \\
 &= 2(\delta_{ib} - p_i) \cdot p_{b|b}(1) \cdot p_b + 2p_{b|b}(1) \cdot p_b \cdot (\delta_{ib} - p_i) \cdot \frac{\phi_1}{1 - \phi_1} \\
 &= 2(\delta_{ib} - p_i) \cdot p_b \cdot p_{b|b}(1) \cdot \frac{1}{1 - \phi_1},
 \end{aligned}$$

and

$$\begin{aligned}
 \sigma_{aa;bb} &= (\delta_{ab} - p_{bb}(1)) \cdot p_{aa}(1) + 2(\delta_{ab} \cdot p_{a|a}(1) \cdot p_{bb}(1) - p_{aa}(1) p_{bb}(1)) \\
 &\quad + \sum_{n=2}^{\infty} (p_{b|b}(1) \cdot p_{b|a}(n-1) \cdot p_{aa}(1) + p_{a|a}(1) \cdot p_{a|b}(n-1) \cdot p_{bb}(1) - 2p_{aa}(1) p_{bb}(1)) \\
 &= (\delta_{ab} - p_{bb}(1)) \cdot p_{aa}(1) + 2p_{a|a}(1) \cdot p_{bb}(1) \cdot (\delta_{ab} - p_a)
 \end{aligned}$$

$$\begin{aligned}
& + p_{b|b}(1) \cdot p_{aa}(1) \cdot \sum_{n=2}^{\infty} (p_{b|a}(n-1) - p_b) + p_{bb}(1) \cdot p_{a|a}(1) \cdot \sum_{n=2}^{\infty} (p_{a|b}(n-1) - p_a) \\
& = (\delta_{ab} - p_{bb}(1)) \cdot p_{aa}(1) + 2p_{a|a}(1) \cdot p_{bb}(1) \cdot (\delta_{ab} - p_a) \\
& \quad + p_{b|b}(1) \cdot p_{aa}(1) \cdot (\delta_{ab} - p_b) \cdot \sum_{n=2}^{\infty} \phi_1^{n-1} + p_{bb}(1) \cdot p_{a|a}(1) \cdot (\delta_{ab} - p_a) \cdot \sum_{n=2}^{\infty} \phi_1^{n-1} \\
& = (\delta_{ab} - p_{aa}(1)) \cdot p_{bb}(1) + \frac{2}{1-\phi_1} \cdot p_{a|a}(1) \cdot p_{bb}(1) \cdot (\delta_{ab} - p_a).
\end{aligned}$$

Finally, for a DAR(1) process, we have $c = \frac{1+\phi_1}{1-\phi_1}$ according to formula (12).

We consider first the estimator $\hat{\kappa}(1)$. From (6), we have

$$\sum_{j=0}^m p_{jj}(1) = \sum_{j=0}^m (p_j^2 \cdot (1-\phi_1) + p_j \cdot \phi_1) = (1-\phi_1) \cdot s_2(\mathbf{p}) + \phi_1.$$

From Theorem 5.1.1, we obtain $T \cdot$ the variance of $\hat{\kappa}(1)$ as

$$\begin{aligned}
& 4c \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1-s_2(\mathbf{p}))^4} \cdot \left(1 - \sum_{j=0}^m p_{jj}(1)\right)^2 - 4 \cdot \frac{1 - \sum_{j=0}^m p_{jj}(1)}{(1-s_2(\mathbf{p}))^3} \cdot \sum_{i,b=0}^m p_i \cdot \sigma_{i;bb} + \frac{\sum_{a,b=0}^m \sigma_{aa;bb}}{(1-s_2(\mathbf{p}))^2} \\
& = 4 \cdot \frac{1+\phi_1}{1-\phi_1} \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1-s_2(\mathbf{p}))^4} \cdot (1-\phi_1)^2 \cdot (1-s_2(\mathbf{p}))^2 - 4 \cdot \frac{1-\phi_1}{(1-s_2(\mathbf{p}))^2} \cdot \frac{2}{1-\phi_1} \cdot \sum_{i,b=0}^m (\delta_{ib} - p_i) \\
& \quad \cdot p_i p_b \cdot p_{b|b}(1) \frac{\sum_{a,b=0}^m (\delta_{ab} - p_{aa}(1)) \cdot p_{bb}(1)}{(1-s_2(\mathbf{p}))^2} + \frac{2}{1-\phi_1} \cdot \frac{\sum_{a,b=0}^m p_{a|a}(1) \cdot p_{bb}(1) \cdot (\delta_{ab} - p_a)}{(1-s_2(\mathbf{p}))^2} \\
& = 4 \cdot (1-\phi_1^2) \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1-s_2(\mathbf{p}))^2} - \frac{8}{(1-s_2(\mathbf{p}))^2} \cdot \sum_{b=0}^m (p_b^3 \cdot (1-\phi_1) + p_b^2 \cdot \phi_1) \\
& \quad + \frac{8}{(1-s_2(\mathbf{p}))^2} \cdot s_2(\mathbf{p}) \cdot \sum_{b=0}^m p_{bb}(1) + \frac{\sum_{a=0}^m p_{aa}(1) - \left(\sum_{a=0}^m p_{aa}(1)\right)^2}{(1-s_2(\mathbf{p}))^2} \\
& \quad + \frac{2}{1-\phi_1} \cdot \frac{\sum_{a=0}^m p_a \cdot p_{a|a}(1)^2}{(1-s_2(\mathbf{p}))^2} - \frac{2}{1-\phi_1} \cdot \frac{\left(\sum_{a=0}^m p_{aa}(1)\right)^2}{(1-s_2(\mathbf{p}))^2} \\
& = 4 \cdot (1-\phi_1^2) \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1-s_2(\mathbf{p}))^2} - 8 \cdot \frac{(1-\phi_1) \cdot s_3(\mathbf{p}) + \phi_1 \cdot s_2(\mathbf{p})}{(1-s_2(\mathbf{p}))^2} + 8 \cdot \frac{(1-\phi_1) \cdot s_2^2(\mathbf{p}) + \phi_1 \cdot s_2(\mathbf{p})}{(1-s_2(\mathbf{p}))^2} \\
& \quad + \frac{(1-\phi_1) \cdot ((1-\phi_1) \cdot s_2(\mathbf{p}) + \phi_1)}{1-s_2(\mathbf{p})} + \frac{2}{1-\phi_1} \cdot \frac{\sum_{a=0}^m p_a \cdot (p_a \cdot (1-\phi_1) + \phi_1)^2}{(1-s_2(\mathbf{p}))^2} \\
& \quad - \frac{2}{1-\phi_1} \cdot \frac{((1-\phi_1) \cdot s_2(\mathbf{p}) + \phi_1)^2}{(1-s_2(\mathbf{p}))^2} \\
& = 4 \cdot (1-\phi_1^2) \cdot \frac{s_3(\mathbf{p}) - s_2^2(\mathbf{p})}{(1-s_2(\mathbf{p}))^2} - 8 \cdot \frac{(1-\phi_1) \cdot (s_3(\mathbf{p}) - s_2^2(\mathbf{p}))}{(1-s_2(\mathbf{p}))^2} + \frac{(1-\phi_1)^2 \cdot s_2(\mathbf{p}) + \phi_1(1-\phi_1)}{1-s_2(\mathbf{p})} \\
& \quad + \frac{2}{1-\phi_1} \cdot \frac{(1-\phi_1)^2 \cdot s_3(\mathbf{p}) + 2\phi_1(1-\phi_1) \cdot s_2(\mathbf{p}) + \phi_1^2}{(1-s_2(\mathbf{p}))^2} \\
& \quad - \frac{2}{1-\phi_1} \cdot \frac{(1-\phi_1)^2 \cdot s_2^2(\mathbf{p}) + 2\phi_1(1-\phi_1) \cdot s_2(\mathbf{p}) + \phi_1^2}{(1-s_2(\mathbf{p}))^2} \\
& = 2 \cdot (2\phi_1 - 1) \cdot \frac{(1-\phi_1) \cdot (s_3(\mathbf{p}) - s_2^2(\mathbf{p}))}{(1-s_2(\mathbf{p}))^2} + \frac{(1-\phi_1) \cdot s_2(\mathbf{p})}{1-s_2(\mathbf{p})} + \phi_1(1-\phi_1).
\end{aligned}$$

Next, we consider the estimator $\hat{\kappa}^*(1)$. We have

$$\begin{aligned}
 & c \cdot \sum_{i=0}^m \frac{p_{ii}(1)^2}{p_i^3} - c \cdot \left(\sum_{i=0}^m p_{ii}(1) \right)^2 - 2 \cdot \sum_{i,b=0}^m \frac{p_{ii}(1)}{p_i^2 p_b} \cdot \sigma_{i;bb} + \sum_{a,b=0}^m \frac{\sigma_{aa;bb}}{p_a p_b} \\
 &= \frac{1+\phi_1}{1-\phi_1} \cdot \sum_{i=0}^m \frac{p_{ii}(1)^2}{p_i} - \frac{1+\phi_1}{1-\phi_1} \cdot \left(\sum_{i=0}^m p_{ii}(1) \right)^2 - \frac{4}{1-\phi_1} \cdot \sum_{i,b=0}^m \frac{(\delta_{ib} - p_i) \cdot p_{ii}(1) \cdot p_{b|b}(1)}{p_i} \\
 &\quad + \sum_{a,b=0}^m \frac{(\delta_{ab} - p_{aa}(1)) \cdot p_{b|b}(1) + \frac{2}{1-\phi_1} \cdot p_{a|a}(1) \cdot p_{b|b}(1) \cdot (\delta_{ab} - p_a)}{p_a} \\
 &= \frac{1+\phi_1}{1-\phi_1} \cdot \sum_{i=0}^m \frac{p_{ii}(1)^2}{p_i} - \frac{1+\phi_1}{1-\phi_1} \cdot \left(\sum_{i=0}^m p_{ii}(1) \right)^2 - \frac{4}{1-\phi_1} \cdot \sum_{i=0}^m \frac{p_{ii}(1)^2}{p_i} + \frac{4}{1-\phi_1} \cdot \left(\sum_{i=0}^m p_{ii}(1) \right)^2 \\
 &\quad + \sum_{a=0}^m \frac{p_{a|a}(1)}{p_a} + \frac{2}{1-\phi_1} \cdot \sum_{a=0}^m \frac{p_{a|a}(1)^2}{p_a} - \left(\sum_{i=0}^m p_{ii}(1) \right)^2 - \frac{2}{1-\phi_1} \cdot \left(\sum_{i=0}^m p_{ii}(1) \right)^2 \\
 &= \left(\frac{1+\phi_1}{1-\phi_1} - \frac{2}{1-\phi_1} \right) \cdot \sum_{i=0}^m \frac{p_{ii}(1)^2}{p_i} + \left(-\frac{1+\phi_1}{1-\phi_1} + \frac{2}{1-\phi_1} - 1 \right) \cdot \left(\sum_{i=0}^m p_{ii}(1) \right)^2 + \sum_{i=0}^m \frac{p_{ii}(1)}{p_i} \\
 &= -\sum_{i=0}^m \frac{p_{ii}(1)^2}{p_i} + 0 + \sum_{i=0}^m \frac{p_{ii}(1)}{p_i} \\
 &= \sum_{i=0}^m \frac{p_{ii}(1)}{p_i} - \sum_{i=0}^m \frac{p_{ii}(1)}{p_i} \cdot (p_i \cdot (1-\phi_1) + \phi_1) \\
 &= (1-\phi_1) \cdot \sum_{i=0}^m p_{ii}(1) \cdot \frac{1-p_i}{p_i} = m \cdot (1-\phi_1)^2 + \phi_1(1-\phi_1) \cdot \sum_{i=0}^m \frac{1-p_i}{p_i}.
 \end{aligned}$$

Appendix D. Proofs of Section 6

D.1. Proof of formulas (24) and (25)

We already know from [Appendix B.1](#) that $\sigma_{ij} = p_j \cdot (\delta_{ij} - p_i) \cdot c$. Next, we consider

$$\begin{aligned}
 E[X_{0;i} \cdot X_{n;ab}(k)] &= E[(Y_{0,i} - p_i) \cdot (Y_{n,a} Y_{n-k,b} - p_{ab}(k))] \\
 &= E[Y_{0,i} Y_{n,a} Y_{n-k,b}] - p_i p_{ab}(k) \\
 &= \begin{cases} \delta_{ia} \cdot p_{ab}(k) - p_i p_{ab}(k) & \text{if } n = 0, \\ p_{aib}(n, k-n) - p_i p_{ab}(k) & \text{if } 1 \leq n \leq k-1, \\ \delta_{ib} \cdot p_{ab}(k) - p_i p_{ab}(k) & \text{if } n = k, \\ p_{abi}(k, n-k) - p_i p_{ab}(k) & \text{if } n \geq k+1, \end{cases}
 \end{aligned}$$

also see [Remark 5.1](#). Analogously,

$$\begin{aligned}
 E[X_{n;i} \cdot X_{0;ab}(k)] &= E[(Y_{n,i} - p_i) \cdot (Y_{0,a} Y_{-k,b} - p_{ab}(k))] \\
 &= E[Y_{n,i} Y_{0,a} Y_{-k,b}] - p_i p_{ab}(k) \\
 &= \begin{cases} \delta_{ia} \cdot p_{ab}(k) - p_i p_{ab}(k) & \text{if } n = 0, \\ p_{iab}(n, k) - p_i p_{ab}(k) & \text{if } n \geq 1. \end{cases}
 \end{aligned}$$

So according to formula (9), we have

$$\begin{aligned}
 \sigma_{i;ab} &= E[X_{0;i} X_{0;ab}(k)] + \sum_{n=1}^{\infty} E[X_{0;i} X_{n;ab}(k)] + \sum_{n=1}^{\infty} E[X_{n;i} X_{0;ab}(k)] \\
 &= (\delta_{ia} - p_i) \cdot p_{ab}(k) + \sum_{n=1}^{k-1} (p_{aib}(n, k-n) - p_i p_{ab}(k)) \\
 &\quad + (\delta_{ib} - p_i) \cdot p_{ab}(k) + \sum_{n=k+1}^{\infty} (p_{abi}(k, n-k) - p_i p_{ab}(k)) + \sum_{n=1}^{\infty} (p_{iab}(n, k) - p_i p_{ab}(k)).
 \end{aligned}$$

If the underlying process $(X_t)_{\mathbb{Z}}$ is i.i.d., then the above joint probabilities factorize into marginal probabilities, such that the whole expression simplifies to

$$\sigma_{i,ab} = (\delta_{ia} - p_i) \cdot p_a p_b + (\delta_{ib} - p_i) \cdot p_a p_b.$$

Finally,

$$\begin{aligned} E[X_{0,ij}(k) \cdot X_{n,ab}(k)] &= E[(Y_{0,i}Y_{-k,j} - p_{ij}(k)) \cdot (Y_{n,a}Y_{n-k,b} - p_{ab}(k))] \\ &= E[Y_{0,i}Y_{-k,j}Y_{n,a}Y_{n-k,b}] - p_{ij}(k)p_{ab}(k) \\ &= \begin{cases} \delta_{ia} \cdot \delta_{jb} \cdot p_{ij}(k) - p_{ij}(k)p_{ab}(k) & \text{if } n = 0, \\ p_{aibj}(n, k - n, n) - p_{ij}(k)p_{ab}(k) & \text{if } 1 \leq n \leq k - 1, \\ \delta_{ib} \cdot p_{aaj}(k, k) - p_{ij}(k)p_{ab}(k) & \text{if } n = k, \\ p_{abij}(k, n - k, k) - p_{ij}(k)p_{ab}(k) & \text{if } n \geq k + 1, \end{cases} \end{aligned}$$

also see Remark 5.1. So according to formula (9), we have

$$\begin{aligned} \sigma_{ij,ab} &= E[X_{0,ij}(k)X_{0,ab}(k)] + \sum_{n=1}^{\infty} (E[X_{0,ij}(k)X_{n,ab}(k)] + E[X_{n,ij}(k)X_{0,ab}(k)]) \\ &= (\delta_{ia} \cdot \delta_{jb} - p_{ab}(k)) \cdot p_{ij}(k) + \sum_{n=1}^{k-1} ((p_{aibj}(n, k - n, n) - p_{ij}(k)p_{ab}(k)) + (p_{iajb}(n, k - n, n) \\ &\quad - p_{ij}(k)p_{ab}(k))) + (\delta_{ib} \cdot p_{aaj}(k, k) - p_{ij}(k)p_{ab}(k)) + (\delta_{ja} \cdot p_{iab}(k, k) - p_{ij}(k)p_{ab}(k)) \\ &\quad + \sum_{n=k+1}^{\infty} ((p_{abij}(k, n - k, k) - p_{ij}(k)p_{ab}(k)) + (p_{ijab}(k, n - k, k) - p_{ij}(k)p_{ab}(k))). \end{aligned}$$

If the underlying process $(X_t)_{\mathbb{Z}}$ is i.i.d., then the above joint probabilities factorize into marginal probabilities, such that the whole expression simplifies to

$$\sigma_{ij,ab} = (\delta_{ia} \cdot \delta_{jb} - p_a p_b) \cdot p_i p_j + (\delta_{ib} - p_b) \cdot p_i p_j p_a + (\delta_{ja} - p_j) \cdot p_i p_a p_b.$$

D.2. Proof of Theorem 6.1.1

For $\mathbf{g}_v$ defined according to formula (27), we obtain

$$\begin{aligned} \frac{\partial}{\partial x_a} \mathbf{g}_{v,ij} &= \frac{\partial}{\partial x_a} \left(\frac{y_{ij}}{(x_i x_j)^{1/2}} - (x_i x_j)^{1/2} \right) = \begin{cases} -1 - \frac{y_{aa}}{x_a^2} & \text{if } a = i = j, \\ -\frac{1}{2} \frac{y_{aj}}{(x_a^3 x_j)^{1/2}} - \frac{1}{2} \left(\frac{x_j}{x_a} \right)^{1/2} & \text{if } a = i \neq j, \\ -\frac{1}{2} \frac{y_{ia}}{(x_i x_a^3)^{1/2}} - \frac{1}{2} \left(\frac{x_i}{x_a} \right)^{1/2} & \text{if } a = j \neq i, \\ 0 & \text{if } a \neq i, j; \end{cases} \\ \frac{\partial}{\partial y_{ab}} \mathbf{g}_{v,ij} &= \frac{\delta_{ia} \delta_{jb}}{\sqrt{x_i x_j}}. \end{aligned}$$

Let $\mathbf{D}_v$ denote the Jacobian of $\mathbf{g}_v$ evaluated in $(p_0, \dots, p_m, p_0^2, p_1 p_0, \dots, p_m^2)$. Then

$$d_{v,ij,a} = \begin{cases} -2 & \text{if } a = i = j, \\ -\left(\frac{p_j}{p_a}\right)^{1/2} & \text{if } a = i \neq j, \\ -\left(\frac{p_i}{p_a}\right)^{1/2} & \text{if } a = j \neq i, \\ 0 & \text{if } a \neq i, j; \end{cases} = \frac{-\delta_{ia} \sqrt{p_j} - \delta_{ja} \sqrt{p_i}}{\sqrt{p_a}}, \quad d_{v,ij,ab} = \frac{\delta_{ia} \delta_{jb}}{\sqrt{p_i p_j}}.$$

Now consider the matrix Σ from formula (24) with the additional simplifications (25) obtained through the i.i.d.-assumption. Then the components of $\Sigma_v := \mathbf{D}_v \Sigma \mathbf{D}_v^\top$ are given by

$$\begin{aligned} (\Sigma_v)_{rs,tu} &= (\mathbf{D}_v)_{rs,\bullet} \Sigma (\mathbf{D}_v)_{tu,\bullet}^\top \\ &= \sum_{i,j=0}^m \sigma_{ij} \cdot d_{v,rs,i} \cdot d_{v,tu,j} + \sum_{i,a,b=0}^m \sigma_{i,ab} \cdot d_{v,rs,i} \cdot d_{v,tu,ab} \end{aligned}$$

$$\begin{aligned}
& + \sum_{j,a,b=0}^m \sigma_{j;ab} \cdot d_{v;rs,ab} \cdot d_{v;tu,j} + \sum_{i,j,a,b=0}^m \sigma_{ij;ab} \cdot d_{v;rs,ij} \cdot d_{v;tu,ab} \\
& = \sum_{i,j=0}^m 1 \cdot p_j \cdot (\delta_{ij} - p_i) \cdot \frac{-\delta_{ri}\sqrt{p_s} - \delta_{si}\sqrt{p_r}}{\sqrt{p_i}} \cdot \frac{-\delta_{tj}\sqrt{p_u} - \delta_{uj}\sqrt{p_t}}{\sqrt{p_j}} \\
& + \sum_{i,a,b=0}^m (\delta_{ia} + \delta_{ib} - 2p_i) \cdot p_a p_b \cdot \frac{-\delta_{ri}\sqrt{p_s} - \delta_{si}\sqrt{p_r}}{\sqrt{p_i}} \cdot \frac{\delta_{ta}\delta_{ub}}{\sqrt{p_a p_b}} \\
& + \sum_{j,a,b=0}^m (\delta_{ja} + \delta_{jb} - 2p_j) \cdot p_a p_b \cdot \frac{\delta_{ra}\delta_{sb}}{\sqrt{p_a p_b}} \cdot \frac{-\delta_{tj}\sqrt{p_u} - \delta_{uj}\sqrt{p_t}}{\sqrt{p_j}} \\
& + \sum_{i,j,a,b=0}^m (\delta_{ia} \cdot \delta_{jb} \cdot p_i p_j + \delta_{ib} \cdot p_i p_j p_a + \delta_{ja} \cdot p_i p_a p_b - 3p_i p_j p_a p_b) \cdot \frac{\delta_{ri}\delta_{sj}}{\sqrt{p_r p_s}} \cdot \frac{\delta_{ta}\delta_{ub}}{\sqrt{p_t p_u}} \\
& = \frac{\sqrt{p_t}}{\sqrt{p_r}} \cdot (\delta_{rt} - p_r) \cdot \sqrt{p_s} \sqrt{p_u} + \frac{\sqrt{p_t}}{\sqrt{p_s}} \cdot (\delta_{st} - p_s) \cdot \sqrt{p_r} \sqrt{p_u} \\
& + \frac{\sqrt{p_u}}{\sqrt{p_r}} \cdot (\delta_{ru} - p_r) \cdot \sqrt{p_s} \sqrt{p_t} + \frac{\sqrt{p_u}}{\sqrt{p_s}} \cdot (\delta_{su} - p_s) \cdot \sqrt{p_r} \sqrt{p_t} \\
& - \frac{(\delta_{rt} + \delta_{ru} - 2p_r) \cdot \sqrt{p_t p_u} \cdot \sqrt{p_s}}{\sqrt{p_r}} - \frac{(\delta_{st} + \delta_{su} - 2p_s) \cdot \sqrt{p_t p_u} \cdot \sqrt{p_r}}{\sqrt{p_s}} \\
& - \frac{(\delta_{tr} + \delta_{ts} - 2p_t) \cdot \sqrt{p_r p_s} \cdot \sqrt{p_u}}{\sqrt{p_t}} - \frac{(\delta_{ur} + \delta_{us} - 2p_u) \cdot \sqrt{p_r p_s} \cdot \sqrt{p_t}}{\sqrt{p_u}} \\
& + \frac{\delta_{rt} \cdot \delta_{su} \cdot p_r p_s + \delta_{ru} \cdot p_r p_s p_t + \delta_{st} \cdot p_r p_t p_u - 3p_r p_s p_t p_u}{\sqrt{p_r p_s p_t p_u}} \\
& = \delta_{rt} \cdot \sqrt{p_s p_u} + \delta_{st} \cdot \sqrt{p_r p_u} + \delta_{ru} \cdot \sqrt{p_s p_t} + \delta_{su} \cdot \sqrt{p_r p_t} - 4\sqrt{p_r p_s p_t p_u} \\
& - \delta_{rt} \cdot \sqrt{p_s p_u} - \delta_{st} \cdot \sqrt{p_r p_u} - \delta_{ru} \cdot \sqrt{p_s p_t} - \delta_{su} \cdot \sqrt{p_r p_t} + 4\sqrt{p_r p_s p_t p_u} \\
& - \delta_{rt} \cdot \sqrt{p_s p_u} - \delta_{st} \cdot \sqrt{p_r p_u} - \delta_{ru} \cdot \sqrt{p_s p_t} - \delta_{su} \cdot \sqrt{p_r p_t} + 4\sqrt{p_r p_s p_t p_u} \\
& + \delta_{rt} \cdot \delta_{su} + \delta_{ru} \cdot \sqrt{p_s p_t} + \delta_{st} \cdot \sqrt{p_r p_u} - 3\sqrt{p_r p_s p_t p_u} \\
& = \delta_{rt} \cdot \delta_{su} - \delta_{rt} \cdot \sqrt{p_s p_u} - \delta_{su} \cdot \sqrt{p_r p_t} + \sqrt{p_r p_s p_t p_u} \\
& = (\delta_{rt} - \sqrt{p_r p_t}) \cdot (\delta_{su} - \sqrt{p_s p_u}).
\end{aligned}$$

Applying the asymptotic results (24), (25) and the delta theorem (Serfling, 1980, Section 3.3), and using that $\mathbf{g}_v(\dots, p_i, \dots, p_i p_j, \dots) = \mathbf{0}$, we obtain that

$$\sqrt{T} \cdot \mathbf{g}_v \left(\underbrace{\dots, \frac{1}{T} \sum_{t=1}^T Y_{t,i}, \dots, \frac{1}{T} \sum_{t=1}^T Y_{t,i} Y_{t-k,j}, \dots}_{=: \mathbf{Y}_t} \right) \xrightarrow{D} N(\mathbf{0}, \boldsymbol{\Sigma}_v).$$

Now we apply the theorem in Section 3.5 of Serfling (1980). We have that $\boldsymbol{\Sigma}_v$ is idempotent with trace m^2 , since

$$\begin{aligned}
(\boldsymbol{\Sigma}_v^2)_{rs,tu} &= \sum_{a,b=0}^m (\boldsymbol{\Sigma}_v)_{rs,ab} \cdot (\boldsymbol{\Sigma}_v)_{ab,tu} \\
&= \sum_{a=0}^m (\delta_{ra} - \sqrt{p_r p_a}) (\delta_{at} - \sqrt{p_a p_t}) \cdot \sum_{b=0}^m (\delta_{sb} - \sqrt{p_s p_b}) (\delta_{bu} - \sqrt{p_b p_u}) \\
&= (\delta_{rt} - \sqrt{p_r p_t}) \cdot (\delta_{su} - \sqrt{p_s p_u}), \\
\text{tr}(\boldsymbol{\Sigma}_v) &= \sum_{r,s=0}^m (\boldsymbol{\Sigma}_v)_{rs,rs} = \sum_{r,s=0}^m (\delta_{rr} - \sqrt{p_r p_r}) (\delta_{ss} - \sqrt{p_s p_s}) = m^2.
\end{aligned}$$

So $T \cdot \mathbf{g}_v(\mathbf{Y}_t)^\top \mathbf{g}_v(\mathbf{Y}_t)$ is asymptotically χ^2 -distributed with m^2 degrees of freedom.

D.3. Proof of Theorem 6.3.1

For $\mathbf{g}_\tau$ defined according to formula (30), we obtain

$$\frac{\partial}{\partial x_a} g_{\tau,ij} = \begin{cases} \frac{-2x_a}{\left(x_a \left(1 - \sum_{r=0}^m x_r^2\right)\right)^{1/2}} - \frac{(y_{aa} - x_a^2) \left(1 - 2x_a^2 - \sum_{r=0}^m x_r^2\right)}{2 \left(x_a \left(1 - \sum_{r=0}^m x_r^2\right)\right)^{3/2}} & \text{if } a = i = j, \\ \frac{-x_j}{\left(x_j \left(1 - \sum_{r=0}^m x_r^2\right)\right)^{1/2}} + \frac{(y_{aj} - x_a x_j) x_a x_j}{\left(x_j \left(1 - \sum_{r=0}^m x_r^2\right)\right)^{3/2}} & \text{if } a = i \neq j, \\ \frac{-x_i}{\left(x_a \left(1 - \sum_{r=0}^m x_r^2\right)\right)^{1/2}} - \frac{(y_{ia} - x_i x_a) \left(1 - 2x_a^2 - \sum_{r=0}^m x_r^2\right)}{2 \left(x_a \left(1 - \sum_{r=0}^m x_r^2\right)\right)^{3/2}} & \text{if } a = j \neq i, \\ \frac{(y_{ij} - x_i x_j) x_a x_j}{\left(x_j \left(1 - \sum_{r=0}^m x_r^2\right)\right)^{3/2}} & \text{if } a \neq i, j; \end{cases}$$

$$\frac{\partial}{\partial y_{ab}} g_{\tau,ij} = \frac{\delta_{ia} \delta_{jb}}{\sqrt{x_j \left(1 - \sum_{r=0}^m x_r^2\right)}}.$$

Let $\mathbf{D}_\tau$ denote the Jacobian of $\mathbf{g}_\tau$ evaluated in $(p_0, \dots, p_m, p_0^2, p_1 p_0, \dots, p_m^2)$. Then

$$d_{\tau,ij,a} = \begin{cases} \frac{-2\sqrt{p_a}}{\sqrt{1-s_2(\mathbf{p})}} & \text{if } a = i = j, \\ \frac{-\sqrt{p_j}}{\sqrt{1-s_2(\mathbf{p})}} & \text{if } a = i \neq j, \\ \frac{-p_i}{\sqrt{p_a(1-s_2(\mathbf{p}))}} & \text{if } a = j \neq i, \\ 0 & \text{if } a \neq i, j; \end{cases} = \sqrt{\frac{p_i}{p_a}} \cdot \frac{-\delta_{ia}\sqrt{p_j} - \delta_{ja}\sqrt{p_i}}{\sqrt{1-s_2(\mathbf{p})}},$$

$$d_{\tau,ij,ab} = \frac{\delta_{ia} \delta_{jb}}{\sqrt{p_j(1-s_2(\mathbf{p}))}}.$$

Now consider the matrix Σ from formula (24) with the additional simplifications (25) obtained through the i.i.d.-assumption. Then the components of $\Sigma_\tau := \mathbf{D}_\tau \Sigma \mathbf{D}_\tau^\top$ are given by

$$\begin{aligned} (\Sigma_\tau)_{rs,tu} &= (\mathbf{D}_\tau)_{rs,\bullet} \Sigma (\mathbf{D}_\tau)_{tu,\bullet}^\top \\ &= \sum_{i,j=0}^m \sigma_{ij} \cdot d_{\tau,rs,i} \cdot d_{\tau,tu,j} + \sum_{i,a,b=0}^m \sigma_{i;ab} \cdot d_{\tau,rs,i} \cdot d_{\tau,tu,ab} \\ &\quad + \sum_{j,a,b=0}^m \sigma_{j;ab} \cdot d_{\tau,rs,ab} \cdot d_{\tau,tu,j} + \sum_{i,j,a,b=0}^m \sigma_{ij;ab} \cdot d_{\tau,rs,ij} \cdot d_{\tau,tu,ab} \\ &= \sum_{i,j=0}^m \frac{p_j \cdot (\delta_{ij} - p_i)}{1 - s_2(\mathbf{p})} \cdot \sqrt{\frac{p_r}{p_i}} \cdot \frac{-\delta_{ri}\sqrt{p_s} - \delta_{si}\sqrt{p_r}}{1} \cdot \sqrt{\frac{p_t}{p_j}} \cdot \frac{-\delta_{tj}\sqrt{p_u} - \delta_{uj}\sqrt{p_t}}{1} \\ &\quad + \sum_{i,a,b=0}^m \frac{(\delta_{ia} + \delta_{ib} - 2p_i) \cdot p_a p_b}{1 - s_2(\mathbf{p})} \cdot \sqrt{\frac{p_r}{p_i}} \cdot \frac{-\delta_{ri}\sqrt{p_s} - \delta_{si}\sqrt{p_r}}{1} \cdot \frac{\delta_{ta} \delta_{ub}}{\sqrt{p_u}} \\ &\quad + \sum_{j,a,b=0}^m \frac{(\delta_{ja} + \delta_{jb} - 2p_j) \cdot p_a p_b}{1 - s_2(\mathbf{p})} \cdot \frac{\delta_{ra} \delta_{sb}}{\sqrt{p_s}} \cdot \sqrt{\frac{p_t}{p_j}} \cdot \frac{-\delta_{tj}\sqrt{p_u} - \delta_{uj}\sqrt{p_t}}{1} \\ &\quad + \sum_{i,j,a,b=0}^m \frac{\delta_{ia} \cdot \delta_{jb} \cdot p_i p_j + \delta_{ib} \cdot p_i p_j p_a + \delta_{ja} \cdot p_i p_a p_b - 3p_i p_j p_a p_b}{1 - s_2(\mathbf{p})} \cdot \frac{\delta_{ri} \delta_{sj}}{\sqrt{p_s}} \cdot \frac{\delta_{ta} \delta_{ub}}{\sqrt{p_u}} \end{aligned}$$

$$\begin{aligned}
&= \frac{\delta_{rt} + \delta_{ru} - 2p_r}{1 - s_2(\mathbf{p})} \cdot p_t \sqrt{p_s p_u} + \frac{\delta_{st} + \delta_{su} - 2p_s}{1 - s_2(\mathbf{p})} \cdot p_r p_t \sqrt{\frac{p_u}{p_s}} \\
&\quad - \frac{\delta_{rt} + \delta_{ru} - 2p_r}{1 - s_2(\mathbf{p})} \cdot p_t \sqrt{p_s p_u} - \frac{\delta_{st} + \delta_{su} - 2p_s}{1 - s_2(\mathbf{p})} \cdot p_r p_t \sqrt{\frac{p_u}{p_s}} \\
&\quad - \frac{\delta_{rt} + \delta_{st} - 2p_t}{1 - s_2(\mathbf{p})} \cdot p_r \sqrt{p_s p_u} - \frac{\delta_{ru} + \delta_{su} - 2p_u}{1 - s_2(\mathbf{p})} \cdot p_r p_t \sqrt{\frac{p_s}{p_u}} \\
&\quad + \frac{\delta_{rt} \cdot \delta_{su} \cdot p_r p_s + \delta_{ru} \cdot p_r p_s p_t + \delta_{st} \cdot p_r p_t p_u - 3p_r p_s p_t p_u}{(1 - s_2(\mathbf{p})) \cdot \sqrt{p_s p_u}} \\
&= 0 + \frac{p_r}{1 - s_2(\mathbf{p})} \cdot (p_t \sqrt{p_s p_u} - \delta_{rt} \cdot \sqrt{p_s p_u} - \delta_{su} \cdot p_t + \delta_{rt} \delta_{su}) \\
&= \frac{p_r}{1 - s_2(\mathbf{p})} \cdot (\delta_{rt} - p_t) (\delta_{su} - \sqrt{p_s p_u}).
\end{aligned}$$

Applying the asymptotic results (24), (25) and the delta theorem (Serfling, 1980, Section 3.3), and using that $\mathbf{g}_r(\dots, p_i, \dots, p_i p_j, \dots) = \mathbf{0}$, we obtain that

$$\sqrt{T} \cdot \mathbf{g}_r \left(\underbrace{\dots, \frac{1}{T} \sum_{t=1}^T Y_{t,i}, \dots, \frac{1}{T} \sum_{t=1}^T Y_{t,i} Y_{t-k,j}, \dots}_{=: \mathbf{Y}_t} \right) \xrightarrow{D} N(\mathbf{0}, \Sigma_r).$$

The trace of Σ_r is equal to m :

$$\text{tr}(\Sigma_r) = \sum_{r,s=0}^m (\Sigma_r)_{rs,rs} = \sum_{r,s=0}^m \frac{p_r(1-p_r)}{1-s_2(\mathbf{p})} \cdot (1-p_s) = m.$$

But the conditions of the theorem in Section 3.5 of Serfling (1980) are not satisfied, i.e., a χ^2 -approximation for $T \cdot \mathbf{g}_r(\mathbf{Y}_t)^\top \mathbf{g}_r(\mathbf{Y}_t)$ is not possible. Applying Theorem 3.1 of Tan (1977), it at least follows that the asymptotic distribution of $T \cdot \mathbf{g}_r(\mathbf{Y}_t)^\top \mathbf{g}_r(\mathbf{Y}_t)$ is the same as the one of $\sum_{i=1}^l \lambda_i \cdot Q_i$, where $\lambda_1, \dots, \lambda_l$ are the nonzero distinct eigenvalues of Σ_r with algebraic multiplicities $r_1, \dots, r_l$, respectively, and where $Q_1, \dots, Q_l$ are independent random variables with Q_i being χ^2 -distributed with r_i degrees of freedom.

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